VNU-HUS MAT1206E/3508: Introduction to AI

Search, Games and Problem Solving

In-class Discussion

Hoàng Anh Đức

Bộ môn Tin học, Khoa Toán-Cơ-Tin học
Đại học KHTN, ĐHQG Hà Nội
hoanganhduc@hus.edu.vn



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- By the 1950s, as computers advanced to the point where experimenting with practical AI algorithms became feasible, *games* emerged as one of the most distinctive AI challenges (aside from tasks like cracking Nazi codes).
- Games offered a *well-defined and constrained domain* that could be *easily formalized and studied*.
- Board games such as *checkers*, *chess*, and more recently the highly complex strategy game *Go* (originating in China over 2500 years ago) have inspired countless researchers and continue to drive advancements in AI.
- Closely tied to games, *search and planning techniques* became a *major area of progress in AI* during the 1960s.
- The roots of *Minimax* go back to 1912; *Alpha-Beta Pruning* was conceived in 1956 and published in 1961 [Russell and Norvig 2021], p. 175. Both remain *foundational for game-playing AI*.
- We will explore the fundamental concepts of *search*, *games*, and *problem-solving*.

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Search Problems

A *search problem* is defined by the following values

State: Description of the world; denoted by s.

Starting state: The initial state.

Goal state: A state satisfying the goal condition.

Actions: The agent's allowed choices.

Solution: Search-tree path from the starting state to a goal state.

Cost function: Costs of action-induced transitions; needed for cost-optimal solutions.

State-space graph: States linked by action-induced transitions.

Search tree: Generated from the start; node n stores $\mathrm{state}(n)$ and path data, so a state may reappear.

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Example 1 (8-Puzzle)

Apply the definition to the 8-puzzle, we get

State: 3×3 matrix S with the values $1, 2, 3, 4, 5, 6, 7, 8$ (once each) and one empty square.

Starting state: An arbitrary state.

Goal state: An arbitrary state.

Actions: Movement of the empty square S_{ij} to the left (if $j \neq 1$), right (if $j \neq 3$), up (if $i \neq 1$), down (if $i \neq 3$).

Solution: The path in the search tree from the starting state to the goal state.

Cost function: The constant function 1, since all actions have equal cost.

State space: The state space is not fully connected: it splits into mutually unreachable components. Therefore, some arbitrary 8-puzzle start/goal pairs are unsolvable.

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Example 2 (8-Puzzle, cont.)

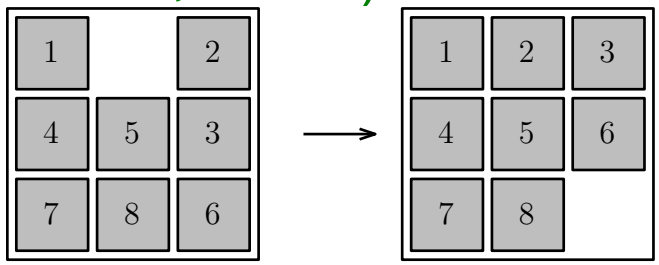


Figure: Possible 8-puzzle start and goal states [Ertel 2025], Fig. 6.2, p. 95

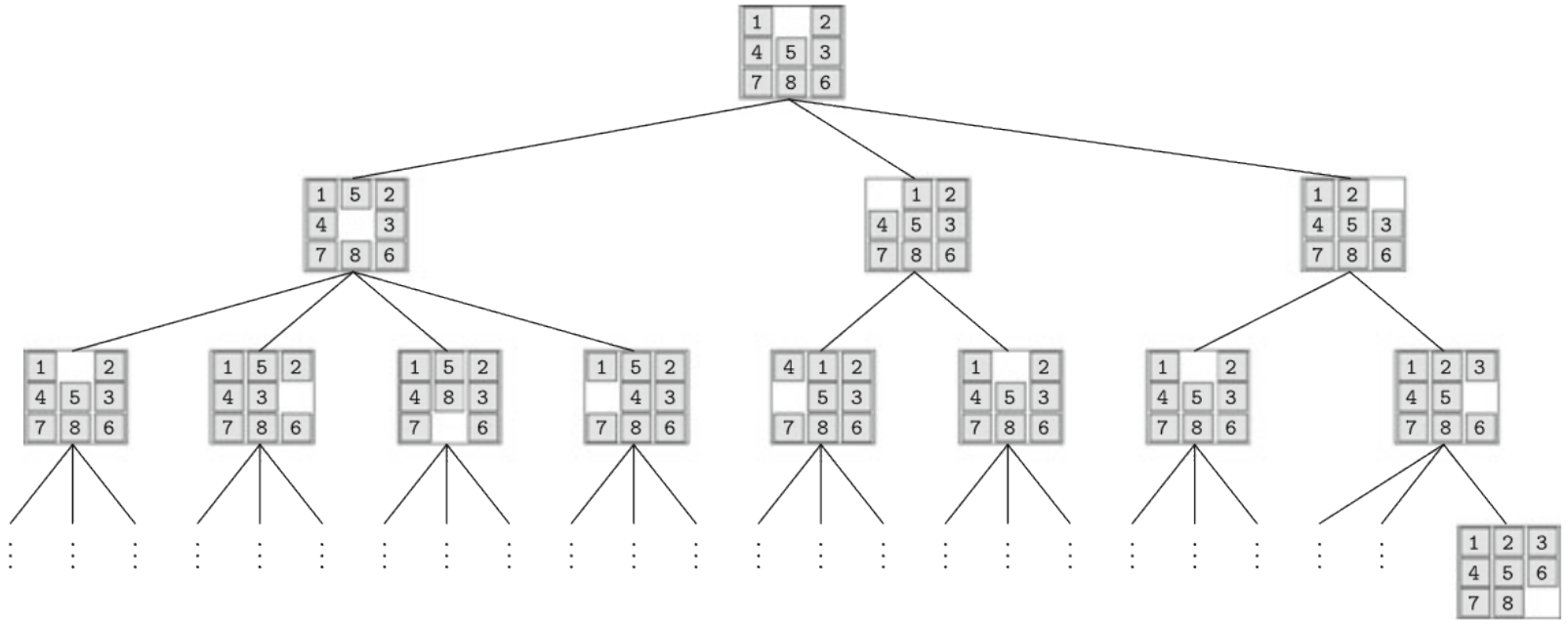


Figure: 8-puzzle search tree; other depth-3 nodes are omitted [Ertel 2025], Fig. 6.3, p. 96



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Branching factor

The *number of successor states of a state* s is called *the branching factor* $b(s)$, or b if *the branching factor is constant*.

Quiz

After applying all legal actions to a state s, we get exactly three successor states. What is $b(s)$?

(a) 0

(b) 1

(c) 3



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Effective branching factor

For a tree of *depth* d with N *total nodes*, the *effective branching factor* is the *constant branching factor* that would give the same d and N.

We count the *root/start node* as depth 0.

Quiz

A tree has depth 3 and 40 total nodes including the root. Which equation defines its effective branching factor $\bar{b}$?

(a) $\bar{b} = 40/3$

(b) $1 + \bar{b} + \bar{b}^2 + \bar{b}^3 = 40$

(c) $\bar{b} = 40$



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Complete Search Algorithms

A search algorithm is *complete* if it *finds a solution whenever one exists*. If complete search exhausts the space with no solution, then *the problem is unsolvable*.

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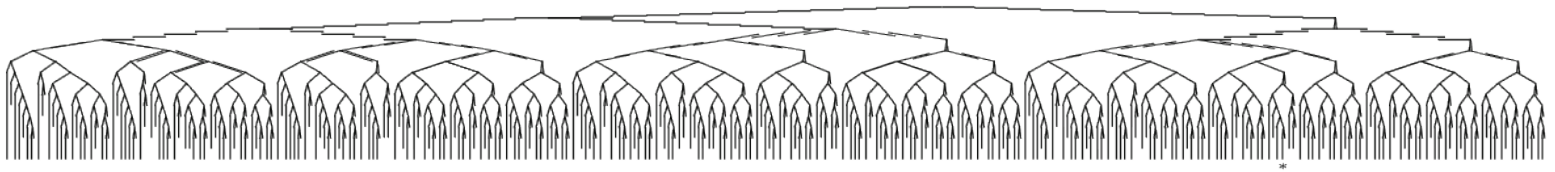
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- The *search* for a solution in an *extremely large search tree* presents a problem for *nearly all inference systems*.
  - From the *starting state* there are *many possibilities for the first inference step*.
  - For each of these possibilities there are again *many possibilities in the next step*, and so on.
- Some reference points for *"large" numbers*:
  - The *age of the universe* is about 1.38×10^{10} years [Planck Collaboration 2020].
  - A *reference adult body* contains on the order of 10^{13} human cells; the estimate depends on the body model and counting method [Hatton et al. 2023].
  - The *human brain* is commonly estimated to contain on the order of 10^{14} synapses [Institute of Medicine 1991].



Example 3

The search tree for SLD resolution proof of *a very simple formula* from [Ertel 1993], as discussed in [Ertel 2025], Sec. 6.1, with *three Horn clauses*, *each with at most three literals*:



- The tree was cut off at a depth of 14 and has a solution in the leaf node marked by $\star$.
- It is only *possible to represent* it at all because of the *small branching factor (= number of children at each node) of at most two* and *a cutoff at depth 14*.



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Quiz

The SLD search tree is cut off at depth 14, the root is at depth 0, and every node has at most 2 children. In the largest possible such tree, how many nodes are at depth 14?

(a) 14

(b) $2^{14} = 16{,}384$

(c) $2^{15} - 1$



Quiz

The displayed SLD tree is cut off at depth 14, the root is at depth 0, and every node has at most 2 children. Why is 2^{14} an upper bound, not an exact count, for the nodes at depth 14?

(a) The branching factor is at most 2, not exactly 2 everywhere.

(b) Depth 14 means there are only 14 nodes.

(c) A solution leaf removes all other branches.



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Quiz

Following [Ertel 2025], Sec. 6.1, assume a full search tree with constant branching factor $b = 30$ is generated up to depth 50, with the root at depth 0. How many nodes are at depth 50?

(a) $30 \cdot 50$

(b) $30^{50} \approx 7.2 \times 10^{73}$

(c) 50^{30}



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Quiz

Assume a full search tree with constant branching factor $b = 30$ is generated up to depth 50, with the root at depth 0. How many total nodes are generated through depth 50, including the root?

(a) 30^{50}

(b) $\sum_{i=0}^{50} 30^i = \frac{30^{51} - 1}{29} \approx 7.4 \times 10^{73}$

(c) $50 \cdot 30^{50}$



Quiz

Assume each generated search-tree node counts as one inference step. With $10{,}000$ computers, each generating 10^9 nodes per second, how long would about 7.4×10^{73} generated nodes take?

(a) 7.4×10^{60} seconds, about 2.3×10^{53} years

(b) 7.4×10^{64} seconds

(c) About 50 years



Quiz

A full search tree with branching factor 30 and depth 50 has about 7.2×10^{73} nodes at depth 50 and about 7.4×10^{73} total generated nodes. What does this illustrate?

(a) For a large constant branching factor, almost all nodes are on the last level.

(b) Inner nodes dominate the computation.

(c) Complete brute-force search is practical for chess.



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Main lesson

If every step creates *many new choices*, the *search tree* becomes *huge very quickly*. So we usually *cannot just try everything*. The rest of the lecture is about how to *search more intelligently*: which nodes to try first, what to remember, and which parts of the tree to avoid.

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Questions

- Why do *good chess players* exist – and nowadays also *good chess computers*?
- Why do mathematicians find proofs for propositions in which the *search space is even larger*?

- Evidently we humans use *intelligent strategies* which *dramatically reduce the search space*.
  - The experienced chess player, just like the experienced mathematician, will, by mere observation of the situation, immediately rule out many actions as senseless. Through his *experience*, he has the *ability to evaluate various actions for their utility in reaching the goal*.
  - Often a person will go by *feel*. If one asks a mathematician how he found a proof, he may answer that the intuition came to him in a dream.
- In everyday problems, *intuition* plays a big role. We will later deal with *heuristic search methods* and with processes by which computers can, similarly to humans, *improve their heuristic search strategies by learning*.

Example 4 (Shortest Path from City A to City B)

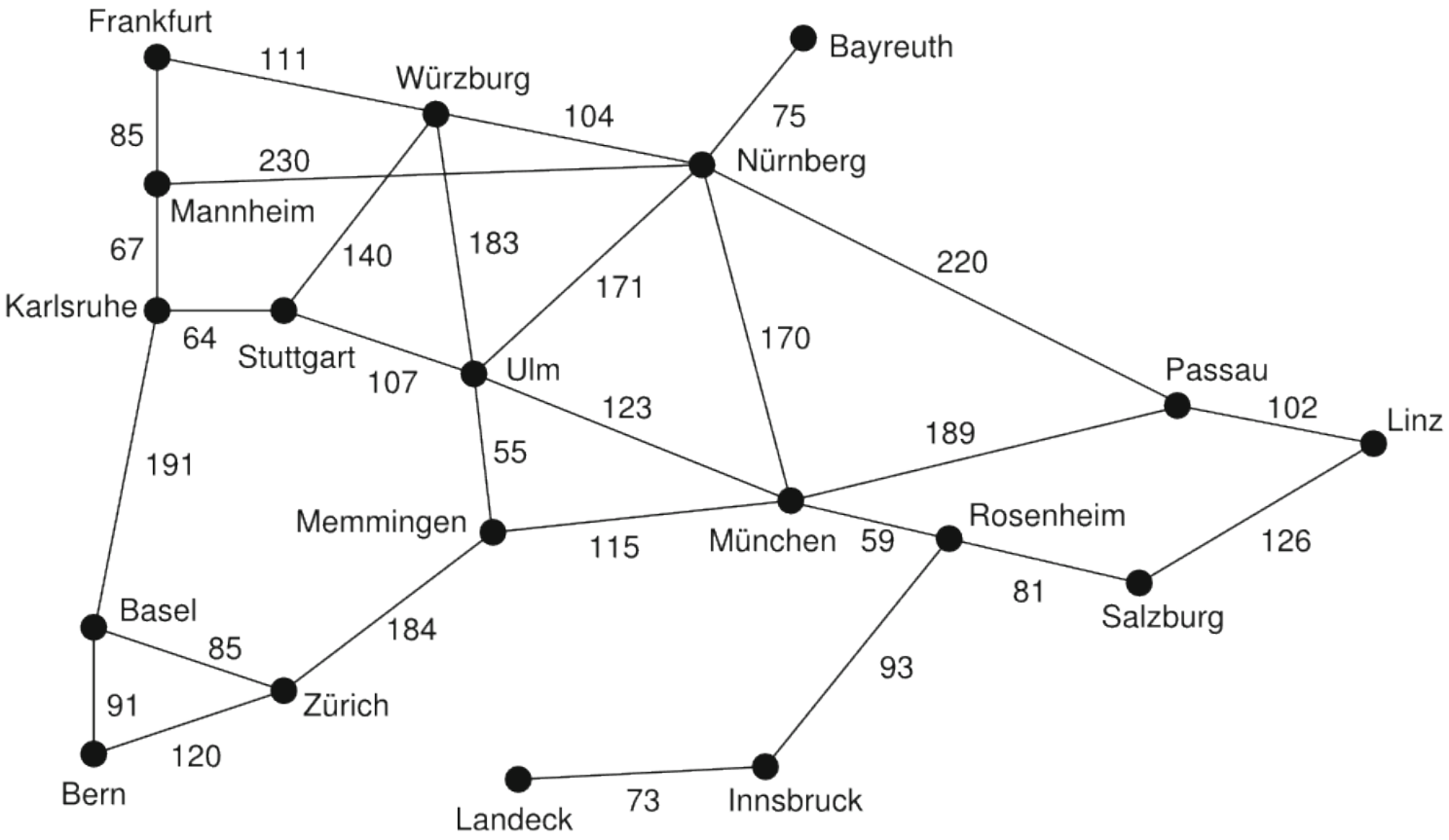


Figure: The graph of southern Germany as an example of a search task with a cost function [Ertel 2025], Fig. 6.5, p. 99.



Example 4 (cont.)

State: A city as the current location of the traveler.

Starting state: An arbitrary city A.

Goal state: An arbitrary city B.

Actions: Travel from the current city to a neighboring city.

Cost function: The distance between the cities. Each action corresponds to an edge in the graph with the distance as the weight.

State space: All cities, represented as vertices of the state-space graph.

Optimal Search Algorithms

A search algorithm is called *optimal* if it *always finds the solution with the lowest cost*, provided that at least one solution exists.

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Deterministic Problem

Every action leads from a state to a unique successor state.

Observable Problem

The agent always knows which state it is in.

Example 5

- The 8-puzzle problem is deterministic and observable.
- In route planning in real applications, one or both characteristics may be missing.
  - The action “Drive from Munich to Ulm” may—for example because of an accident—lead to the successor state “Munich”.
  - It can also occur that the traveler no longer knows where he is because he got lost.

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- For the search algorithms considered next, we assume a *single-agent problem* that is *deterministic and observable*, and whose *action outcomes are known in advance* [Russell and Norvig 2021], Sec. 2.3.2.
- For the 8-puzzle, this known abstract model lets us *find an action sequence before carrying out the moves*. $\Rightarrow$ Offline algorithms.
- Robot soccer is different: an *exact abstract model of the actions* is not available, and the robot must *choose actions from sensor signals while acting*. $\Rightarrow$ Online algorithms. (*Reinforcement learning* works toward optimization of these decisions based on experience.)

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- *Uninformed search* uses only the information given in the search problem. It has *no additional guidance* telling it which current state appears closer to a goal.
- This *does not mean random search*: each method follows a general rule for choosing which node to explore next.

Uninformed Search

Example Problem: Simplified Sliding-Block Puzzle



Toy marker-grid version: one marker moves; the blocked cell cannot be entered.

Initial position

A	#	B
C	D	E
F	G	H

↓ move down

Next position

A	#	B
C	D	E
F	G	H

State Marker location in one cell $A, \ldots, H$.

Starting state B.

Goal state Marker in G.

Actions Move the marker one cell left, right, up, or down when the destination is inside the grid and not blocked.

Solution A legal sequence of moves from B to G.

Cost function Cost 1 for each legal move.

State-space graph Vertices $A, \ldots, H$ linked by legal moves.

Search tree Root B; each child represents one legal move from its parent.

Note. A state is the whole board, but only the marker's cell changes.

That cell therefore identifies the state and labels each search-tree node.

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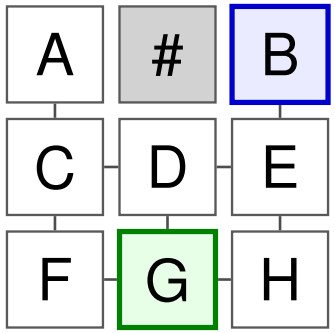
BFS explores the search tree *level by level*: it tests every node at depth i before any node at depth $i + 1$.

- BREADTHFIRSTSEARCH(NodeList, Goal) receives the current level in left-to-right order and starts with NewNodes = $\emptyset$.
- For each Node in NodeList:
  - if GOALREACHED(Node, Goal) finds that state(Node) satisfies the goal, return the solution;
  - otherwise append the child search nodes returned by SUCCESSORS(Node) to NewNodes.
- After the current level is processed, call BREADTHFIRSTSEARCH(NewNodes, Goal). Thus generated successors are tested only in the next invocation.
- If NewNodes = $\emptyset$, return “No solution.”

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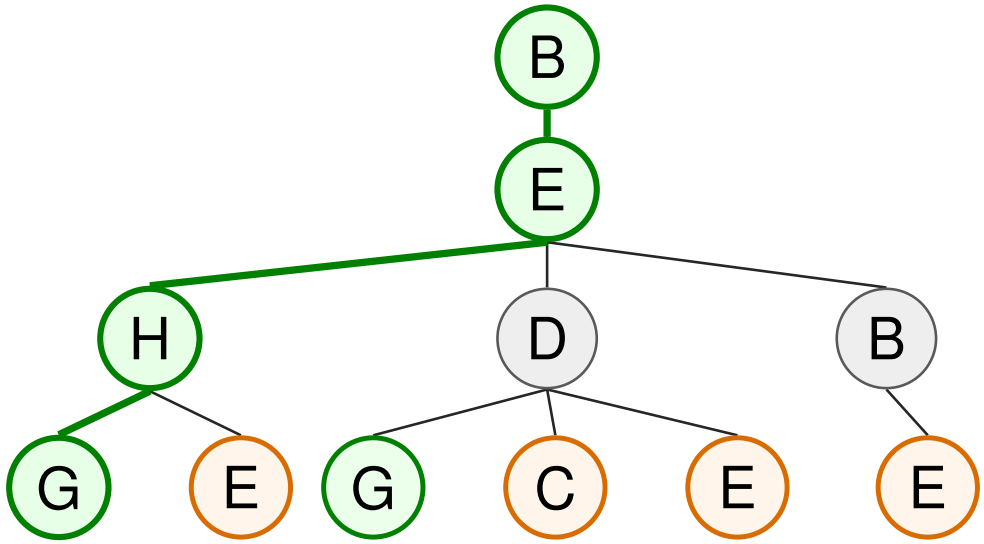
Simplified Sliding-Block Puzzle Solved by BFS

State-space graph



Search tree

Order: down, left, right, up



NodeList	G	E	G	C	E	E

NewNodes: goal found; NewNodes is not used

Following parent links gives $B \to E \to H \to G$, with cost 3. The G generated from D is never tested.

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Breadth-First Search



Characteristics: b: branching factor; d: shallowest goal depth.

- *Complete* if *each node has only finitely many successors*.
- *Optimality is guaranteed* only when *all step costs are equal*.
- *Worst-case time and space:* The algorithm goal-tests $O(b^d)$ nodes but may generate $O(b^{d+1})$ successors, requiring $O(b^{d+1})$ time. The next-level list may store $O(b^{d+1})$ nodes, requiring $O(b^{d+1})$ space. For fixed $b > 1$, both bounds can be written as $O(b^d)$.

When *step costs are not all equal*, we use a variant of BFS.

Uniform-Cost Search

Select an open node with the *smallest path cost from the start*, and test for a goal *when that node is selected*. With finite branching and every action cost at least *some fixed positive amount*, UCS is complete and returns a lowest-cost solution.

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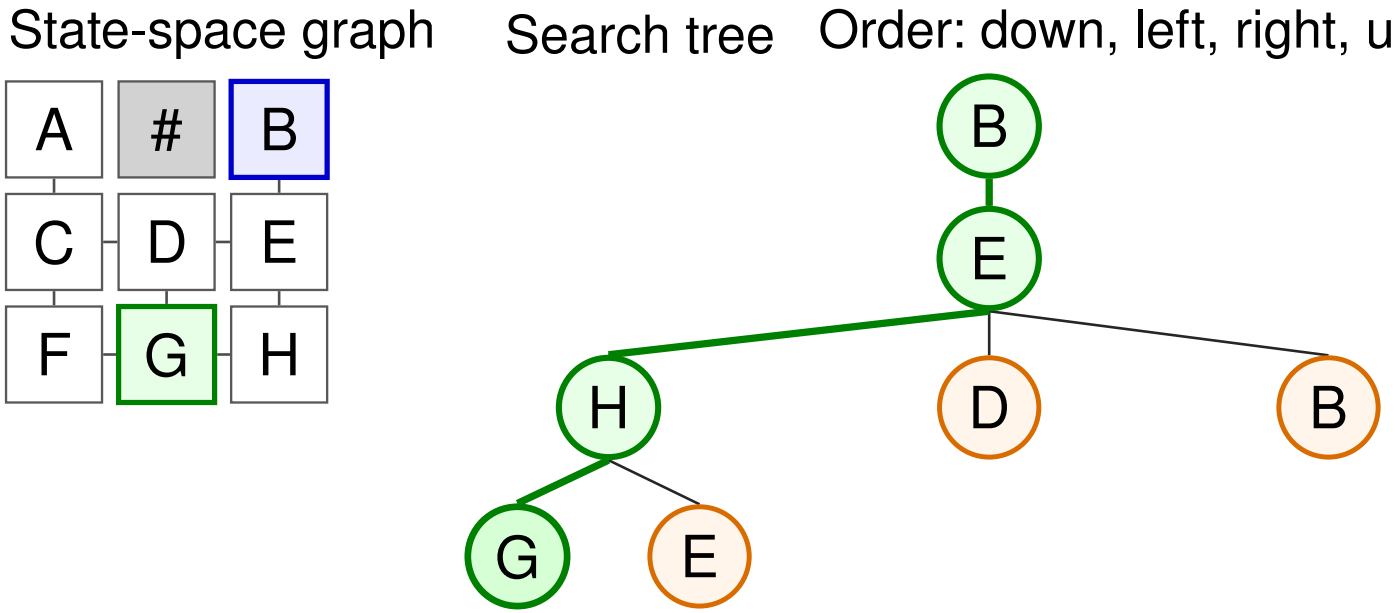
Core Idea

DFS follows *one path as deeply as possible*. If that path has no goal, it *backtracks to the most recent node with an untried successor*.

- DEPTHFIRSTSEARCH(Node, Goal) first uses GOALREACHED(Node, Goal).
- If Node is not the goal, NewNodes receives SUCCESSORS(Node).
- Recursively search FIRST(NewNodes). If it succeeds, return the solution; otherwise replace NewNodes by REST(NewNodes) and try the next successor.
- If every successor fails, return “No solution.”

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Simplified Sliding-Block Puzzle Solved by DFS



Current path B E H G

SUCCESSORS GOALREACHED(G, G) succeeds

GOALREACHED(G,G) succeeds. The recursive call chain gives $B \to E \to H \to G$, with cost 3.

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Characteristics:

- *Incomplete: May follow an infinite branch forever*.
  - For this counterexample, change the move order to left, right, up, down. Without a repeated-state check, DFS follows $B \to E \to D \to C \to D \to C \to \cdots$.
  - Yet $B \to E \to D \to G$ is a solution.
- *Non-optimal:* The first solution found *may be deeper or more costly than another solution*.
- *Time Complexity:* $O(b^{d_s})$ for a finite search tree, where b is the branching factor and d_s is the tree's maximum depth.
- *Space Complexity:* $O(bd_s)$; stores only the current path and its successors.

Depth-Limited Search

A variant of DFS with a depth limit ($\Rightarrow$ *Pruned search tree*). It avoids infinite loops but may miss solutions beyond the limit.

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Iterative Deepening



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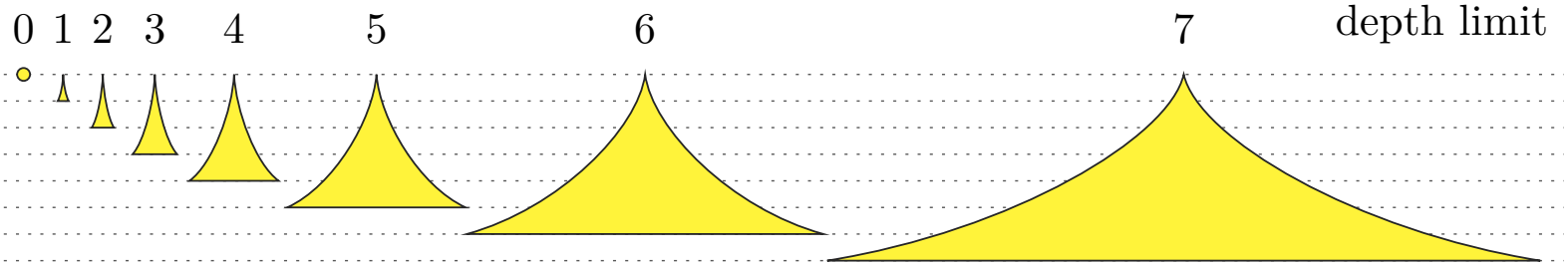
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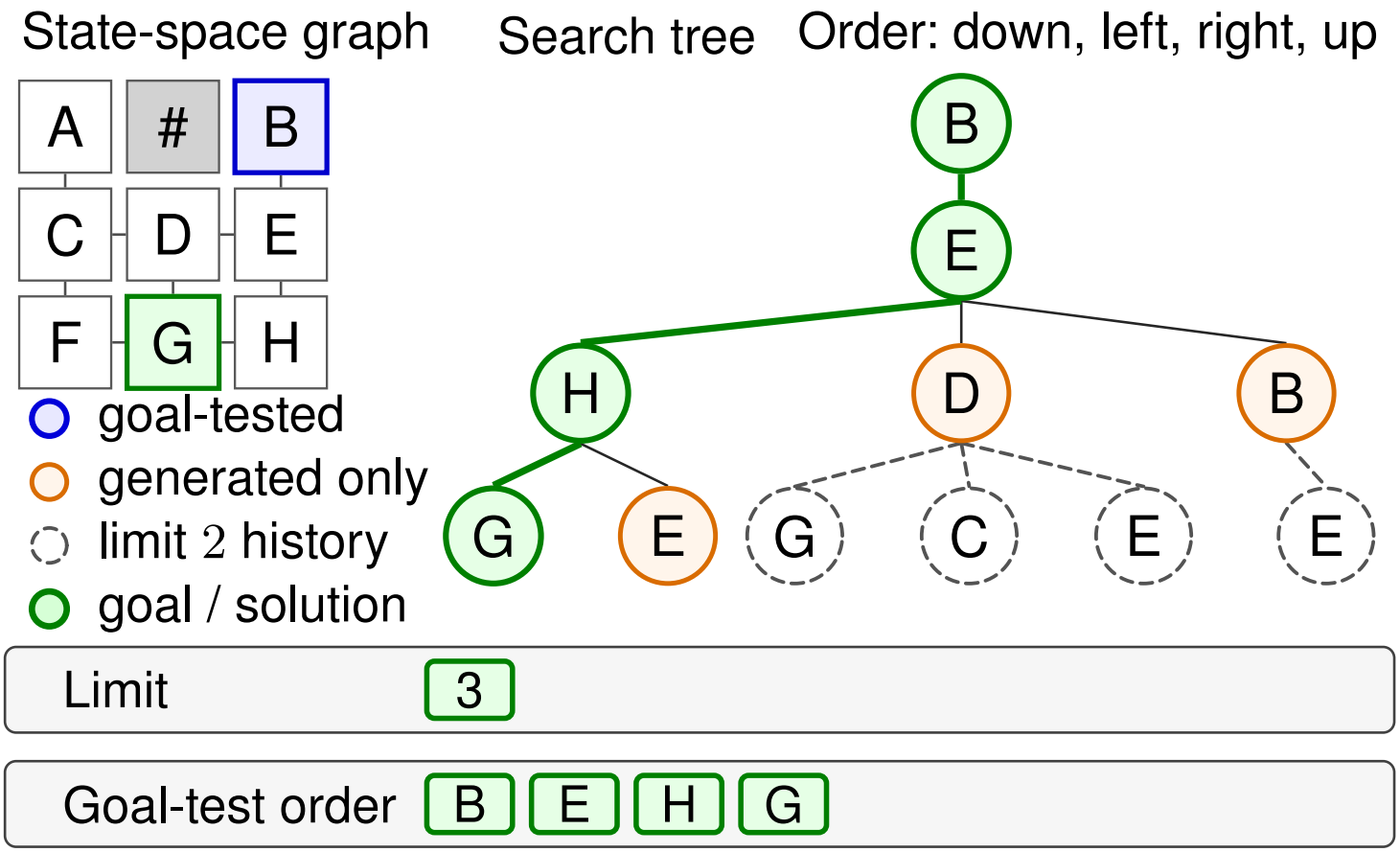
Iterative deepening repeats *depth-limited DFS* with *Limit* $= 0, 1, 2, \ldots$. Each iteration *restarts at the starting node* and *stops if it finds a goal*.

- *Depth* is the current node's depth, with the root at depth 0; *Limit* is the greatest depth searched in the current iteration.
- A reached node is *goal-tested*. If *Depth* $<$ *Limit*, its successors are *searched recursively in DFS order*.
- If *Depth* $=$ *Limit*, successors may be generated, but they are *not recursively visited or goal-tested in that iteration*.



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Simplified Sliding-Block Puzzle Solved by Iterative Deepening



The successful recursion path is $B \to E \to H \to G$. Three unit-cost moves give cost 3. Limits 0–2 goal-tested every node through depth 2, so no shallower solution exists.

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Analysis: Let b be the branching factor and d the shallowest goal depth.

- *Complete.*
- *Optimal* if all action costs are equal and the limit increases by 1.
- *Worst-case time:* $O(b^{d+1})$ for the displayed procedure. It goal-tests $O(b^d)$ nodes but may generate $O(b^{d+1})$ successors because it still calls SUCCESSORS at the limit.
- *Memory requirement:* $O(bd)$; the generated fringe is not stored all at once.
- Restarting repeats work, but for large b the *last iteration dominates the geometric sum*. **[Why?]**

Uninformed Search
Comparison



	Breadth-first search	Uniform-cost search (‡)	Depth-first search	Iterative deepening
Completeness	Yes	Yes	No	Yes
Optimal solution	Yes (*)	Yes	No	Yes (*)
Computation time	b^{d+1} (†)	$b^{1+\lfloor C^*/\epsilon \rfloor}$	∞ or b^{d_s}	b^{d+1} (†)
Memory use	b^{d+1}	$b^{1+\lfloor C^*/\epsilon \rfloor}$	bd_s	bd

- (*): only with constant action cost. (‡): for UCS, finite branching and costs of at least a fixed $\epsilon > 0$; goal-test on selection.
- (†): these late-test procedures test $O(b^d)$ nodes but can generate $O(b^{d+1})$ successors.
- C^*: lowest solution cost; d: shallowest goal depth; d_s: maximal finite-tree depth.

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Cycle Check: What Is Being Checked?



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- A *cycle* occurs when the *current path reaches a state that already appears on that same path*.
  - In the 8-puzzle, one move can often be immediately undone, creating a cycle of length 2.
- A *full cycle check* is a *path check*: when a successor is generated, compare its state with *all predecessor states on the current path*.



Quiz

For a full cycle check, each search-tree node stores all predecessor states on its current path. When a successor is newly generated at depth i, which states should it be compared with?

(a) All nodes generated earlier by the whole search.

(b) The i predecessor states on its current path.

(c) Only the root node.

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Quiz

Assume a full search tree with constant branching factor b. At depth i, there are b^i generated nodes. For a full cycle check, each generated node at depth i is compared only with the i predecessor states on its current path. How many predecessor-state comparisons are made for all nodes at depth i?

(a) i

(b) b^i

(c) $i \cdot b^i$

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Quiz

Assume a full search tree with constant branching factor b is generated through depth d. Generating one node costs c_1. In the full cycle check, each node at depth i is compared with its i path predecessors, and one such comparison costs c_2. Which expression counts the generation cost plus the full cycle-check comparison cost?

(a) $c_1 \sum_{i=0}^{d} b^i$

(b) $c_2 \sum_{i=0}^{d} i \cdot b^i$

(c) $c_1 \sum_{i=0}^{d} b^i + c_2 \sum_{i=0}^{d} i \cdot b^i$



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Quiz

Assume a full search tree with constant branching factor b is generated through depth d. Without the full cycle check, the generation work is about $\sum_{i=0}^{d} b^i$, dominated by b^d for large b and d. With the full cycle check, depth i adds $i \cdot b^i$ predecessor comparisons. What is the main effect on the time complexity?

(a) The extra work is constant.

(b) The extra work is about a factor d, because $\sum_{i=0}^{d} i \cdot b^i$ is dominated by $d \cdot b^d$.

(c) The check compares each node with all earlier generated nodes.

Heuristic Search
Core Idea



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Core Idea

Heuristic search keeps NodeList, a list of *generated but not yet expanded nodes*, ordered by *increasing evaluation* $f(n)$.

- *Initialize:* set NodeList to $[n_0]$, the root node for the start.
- *Select:* remove the first node from NodeList; it has the *smallest evaluation*.
- *Test:* if this node is the goal, *return the solution*.
- *Expand:* otherwise generate its successors and *insert them into NodeList in increasing evaluation order*.
- *Failure:* if NodeList becomes empty, *no solution was found*.

Why the Evaluation Function Matters

The *definition of f determines the search order*: different choices of f give different search methods.

Heuristic Search

Ideally, the *best heuristic* would calculate the actual remaining cost from $\mathrm{state}(n)$ to a goal.

Question

How do we find a heuristic that is fast and simple to compute?

An idea for finding a heuristic *simplification of the problem*.

- The *original task* is *simplified enough* that *it can be solved with little computational cost*.
- The *costs from a state to the goal in the simplified problem* then *serve as an estimate for the actual problem*.
  $\Rightarrow$ *cost estimate function* $h(n)$, computed from $\mathrm{state}(n)$.

Notation

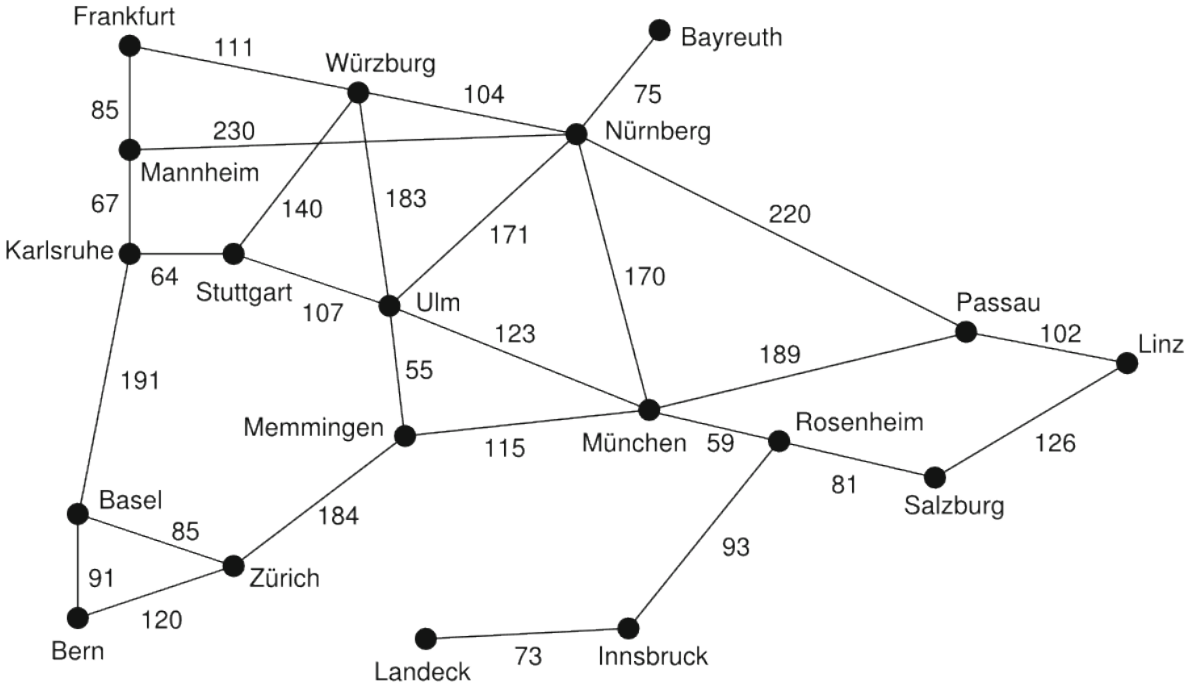
Ertel sometimes uses s for both states and search nodes. Here s always denotes a state and n a search-tree node [Ertel 2025], pp. 106, 109.

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Example 6 (Shortest Path from City A to City B)

- **Simplified Task:** finding the straight line path from city to city (that is, the flying distance).
- *Cost estimate function* $h(n)$ = flying distance from the city stored in node n to Ulm (given in the figure below).



Basel	204
Bayreuth	207
Bern	247
Frankfurt	215
Innsbruck	163
Karlsruhe	137
Landeck	143
Linz	318
München	120
Mannheim	164
Memmingen	47
Nürnberg	132
Passau	257
Rosenheim	168
Stuttgart	75
Salzburg	236
Würzburg	153
Zürich	157

Figure: Flying distances to Ulm [Ertel 2025], Fig. 6.14, p. 108

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Example 6 (cont.)

Greedy search uses the cost estimate function directly: $f(n) = h(n)$.

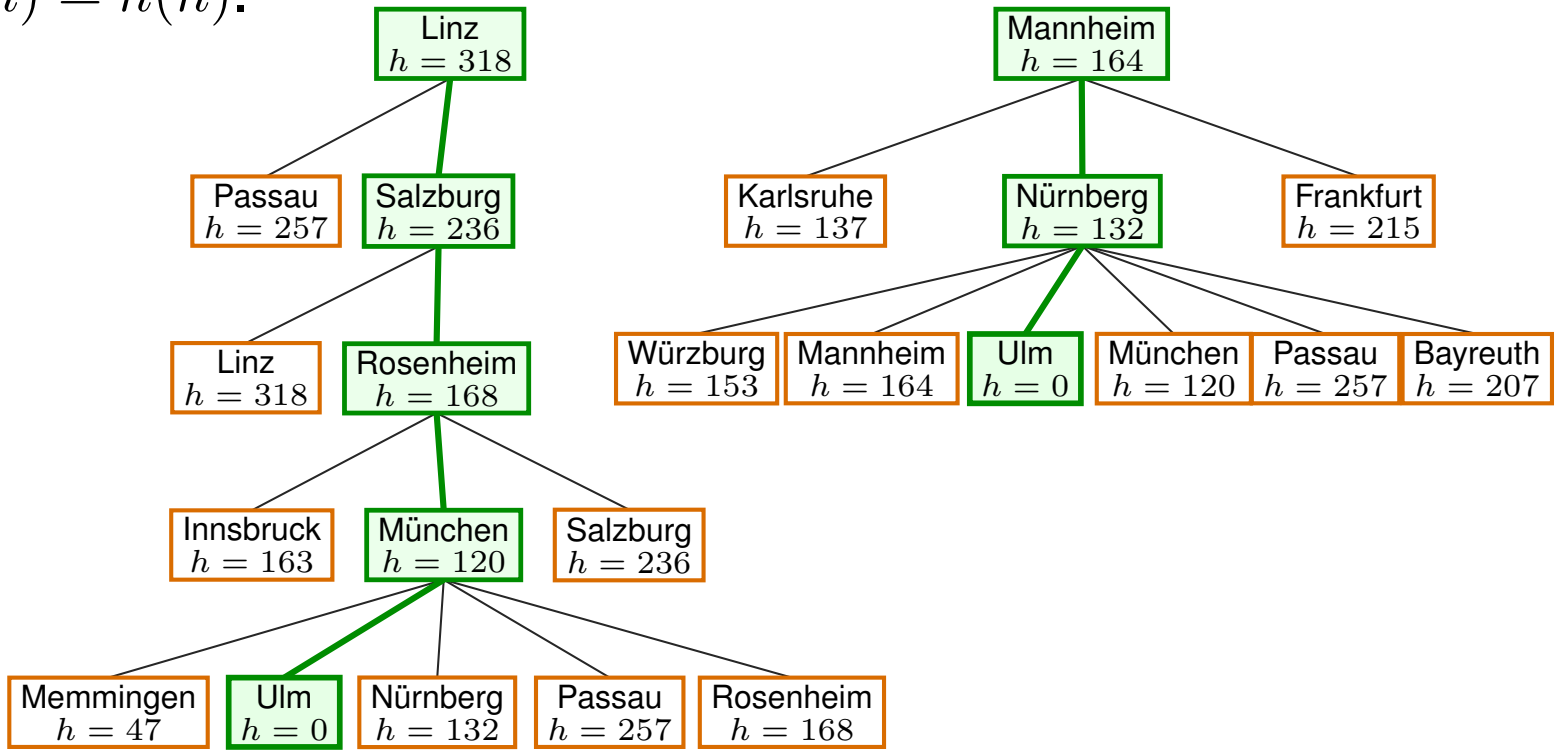


Figure: Select Ulm. Greedy: Mannheim–Nürnberg–Ulm, $230 + 171 = 401$ km. Shorter: Mannheim–Karlsruhe–Stuttgart–Ulm, $67 + 64 + 107 = 238$ km. Thus greedy search is *not always optimal*.

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Quiz

Start at S; only G is a goal. The only moves are $S \to A$ (cost 1), $S \to B$ (cost 6), $A \to G$ (cost 4), and $B \to G$ (cost 1). The estimates are $h(A) = 3$, $h(B) = 1$, and $h(G) = 0$. Greedy search keeps NodeList sorted by h. What is the cost of the route that greedy search returns?

(a) 5

(b) 6

(c) 7

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- *Path-cost function* $g(n)$ = Cost of the path from the root to node n.
- *Heuristic cost estimate* $h(n)$ = Estimated cost from $\text{state}(n)$ to a goal.
- *Heuristic evaluation function* $f(n) = g(n) + h(n)$.

Admissible heuristic cost estimate function

A heuristic $h(n)$ is *admissible* if it is nonnegative and *never overestimates the lowest actual cost* from $\text{state}(n)$ to a goal. Nodes with the same state share h; their g and f values may differ.

A*-algorithm

= HEURISTICSEARCH + $f(n) = g(n) + h(n)$ + admissible h



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Quiz

A* starts in Frankfurt and uses h = flying distance to Ulm. Node n stores Karlsruhe, reached by Frankfurt–Mannheim (85 km) and Mannheim–Karlsruhe (67 km), and $h(n) = 137$ km. What is $f(n) = g(n) + h(n)$?

(a) 137

(b) 204

(c) 289



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Quiz

Suppose NodeList holds three nodes with path cost g and estimate h: n_A ($g = 8$, $h = 1$), n_B ($g = 2$, $h = 4$), and n_C ($g = 4$, $h = 3$). Which node is selected next if NodeList is sorted by $f(n) = h(n)$ (greedy search), and which if it is sorted by $f(n) = g(n) + h(n)$ (A*)?

(a) Greedy: n_A; A*: n_B

(b) Greedy: n_A; A*: n_A

(c) Greedy: n_B; A*: n_C

Heuristic Search
Quiz: Admissible Estimates



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Admissible heuristic

A heuristic h is *admissible* if it is nonnegative and *never overestimates the lowest actual cost* to a goal.

Quiz

A problem has only the states s_A, s_B, and the goal z. Their lowest actual costs to the goal are 5, 2, and 0. Which estimates $\big(h(s_A), h(s_B), h(z)\big)$ are admissible?

(a) $(6, 1, 0)$

(b) $(4, 2, 1)$

(c) $(4, 2, 0)$

Example 6 (cont.)

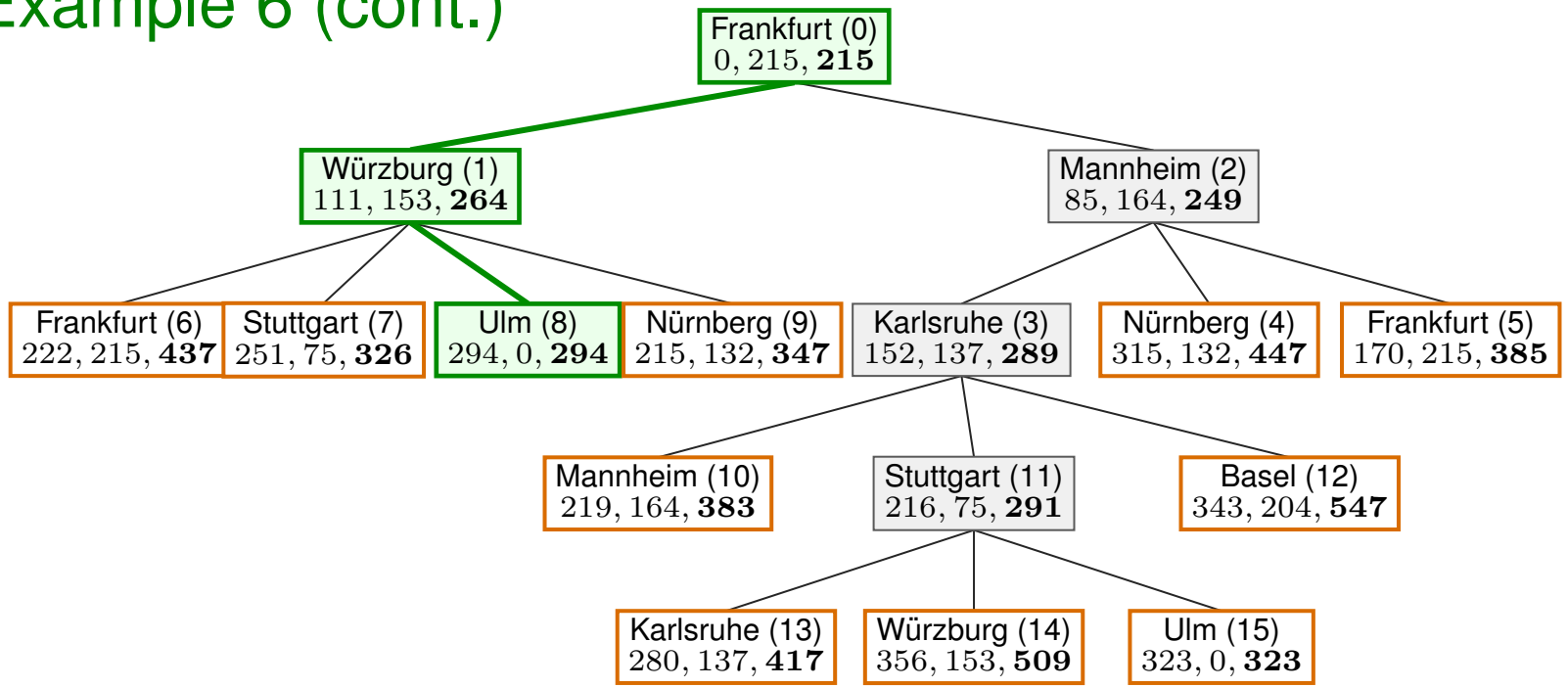


Figure: Select Ulm(8). A* returns Frankfurt–Würzburg–Ulm with total cost 294.

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Quiz

Start at S; only G is a goal. The only moves are $S \to G$ (cost 10), $S \to A$ (cost 2), $A \to B$ (cost 1), and $B \to G$ (cost 2). The estimates are $h(S) = 4$, $h(A) = 2$, $h(B) = 1$, and $h(G) = 0$. What is the cost of the route that A* returns?

(a) 4

(b) 5

(c) 10

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The following restates Proposition 3.1 of Poole and Mackworth in the notation and terminology of these slides [Poole and Mackworth 2023], Sec. 3.6.1.

Theorem 1 (Completeness and cost optimality)

For A search, suppose that the branching factor is bounded by a finite number, every action cost is greater than some fixed* $\epsilon > 0$*, and* h *is admissible. If a solution exists, A* eventually selects a goal, and the first goal selected represents a minimum-cost solution.*

This is the *tree-search form* (different paths to the same state remain separate nodes; *graph search* instead detects repeated states and retains the lowest-cost path found so far for each state) of the original A* *admissibility result* of Hart, Nilsson, and Raphael [Hart, Nilsson, and Raphael 1968], Thm. 1, pp. 102–103.



- Weaknesses of A*:
  - High memory requirements
- Solution: IDA*-algorithm [Korf 1985]
  - Iterative Deepening
  - The same as depth-first search, but limit for heuristic evaluation $f(n)$ (instead of depth limit).
    - Denote the current threshold by θ.
    - Initially, set $\theta = f(n_0)$ for the root node n_0.
    - In one DFS iteration, expand a node n only if $f(n) \leq \theta$; cut off its branch for that iteration when $f(n) > \theta$.
    - Stop as soon as a goal node is reached.
    - If no goal is found, set the next θ to the minimum exceeded f value; if there is no such value, report failure.

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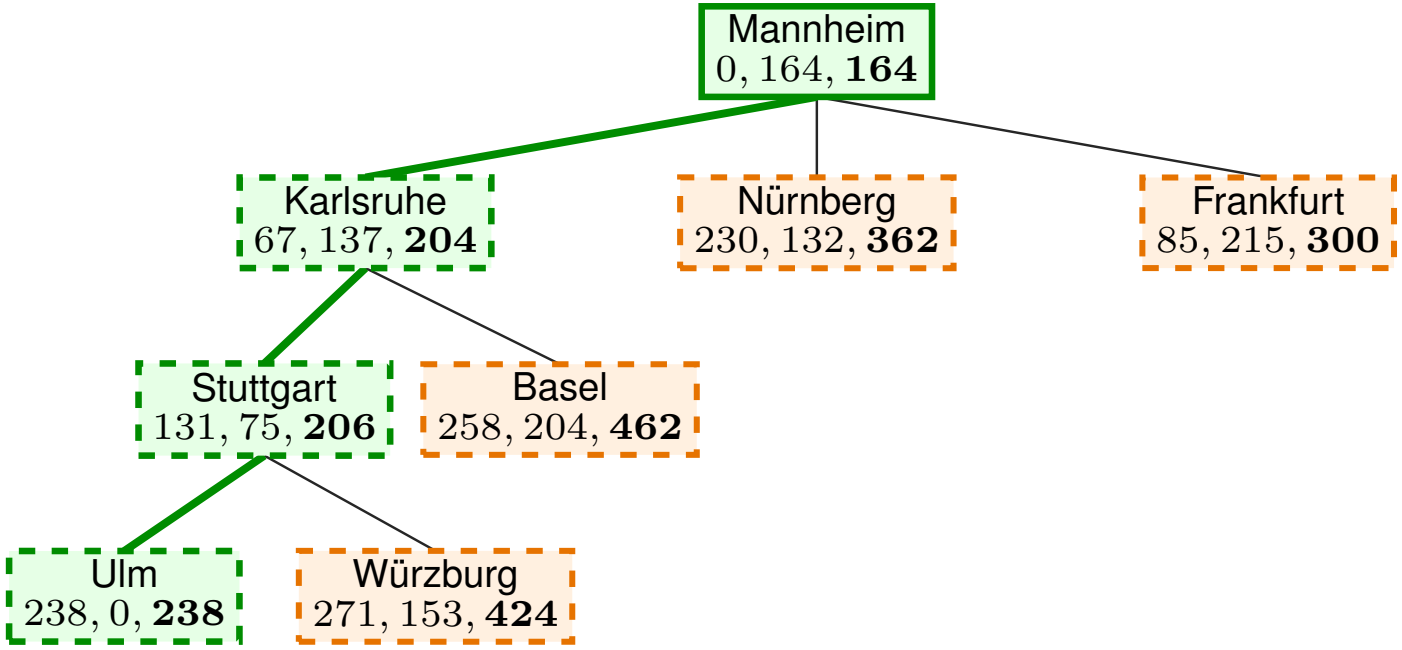
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Example 7 (Route from Mannheim to Ulm)

We use IDA* to find a shortest Mannheim–Ulm route. Edge costs are road distances; h is flying distance to the goal. Labels: $g, h, \mathbf{f} = g + h$; at Mannheim, $\theta = f(n_0) = 164$. Search left to right; do not revisit a city already on the current path.



Iteration 4. With $\theta = 238$, expand Mannheim, Karlsruhe, and Stuttgart. Generated f-value $238 \leq \theta$ reaches Ulm, so no next θ is needed; route Mannheim–Karlsruhe–Stuttgart–Ulm costs 238 km.

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Quiz

An IDA* iteration uses the threshold $\theta = 5$. It expands the root and cuts off the root's three children, whose f-values are 7, 12, and 9. No goal is found. What is θ in the next iteration?

(a) 6

(b) 7

(c) 12

Heuristic Search
Summary



- Among the *uninformed algorithms compared here*, *iterative deepening* is *complete* and uses *very little memory*. Even it may be impractical when the search space is extremely large.
- *IDA** applies successive f*-cost thresholds* using depth-first memory. It saves memory compared with A* but may *regenerate nodes* across iterations.
- *Good heuristics* greatly *reduce the effective branching factor*.
- If a heuristic *only changes the exploration order*, proving that no solution exists still requires *exhausting the relevant reachable search space*.
  - For arbitrary first-order formulas, deciding *logical validity* is undecidable. An unsuccessful proof search may therefore *continue indefinitely*.

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Quiz: Which Kind of Game?



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Properties of Games

Deterministic: the next position depends only on the current position and the move.
Observable: every player always knows the whole game state.

Quiz

In backgammon, both players always see the whole board, but the next position also depends on a dice roll. How do we classify backgammon?

(a) Deterministic and observable.

(b) Not deterministic, but observable.

(c) Deterministic, but not observable.

Games with Opponents

What the Algorithms Solve



Goal: Choose the *next move* that *performs best against the opponent's best replies* [Ertel 2025], Sec. 6.4.1.

Input: Current position, player to move, legal moves and resulting positions, and leaf scores [Russell and Norvig 2021], Sec. 5.1.1.

Output: A *recommended next move*.

Why important? The opponent *chooses against us*, so we must *consider their replies*.

Algorithms

- *Minimax search* assumes the opponent's best replies and *chooses our best move*.
- *Alpha-beta pruning* preserves minimax's *root value* and returns *an optimal move*, while skipping subtrees that cannot change that value.

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Minimax Search



- A typical chess position has *about* 30 *to* 35 *legal moves*. Using $b = 30$ as a simple model, 50 moves per player gives $30^{100} \approx 10^{148}$ leaves, so *full search is infeasible*.
- A *time limit* means that search must *stop at a chosen depth*.
- Minimax first assigns values to positions where its look-ahead stops:
  - if the *game has finished*, use its exact outcome score;
  - if the game is unfinished but the *search limit has been reached*, estimate its value using $B(s)$.
    - **Why?** Small games may be searched to the end. In large games, searching every continuation is infeasible, so a depth or time limit may stop the search earlier.
- Call the player whose result we optimize *Max* and the opponent *Min*. Assume Min *always chooses their best move*.
- At an internal node, calculate its *minimax value* from its children: Max takes the highest value and Min the lowest. All values are from *Max's perspective*.

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Minimax Search: Tic-Tac-Toe Example



- **Game Setup:** A 3x3 Tic-Tac-Toe board with two players: Max (X) and Min (O). Players alternate turns, with Max going first.
- **Objective:** The exact score of a finished game is $+1$ for a Max win, -1 for a Min win, and 0 for a draw. A player wins by placing three of their marks in a horizontal, vertical, or diagonal row.
- **Minimax Algorithm:**
  - Starting from the leaf scores, calculate each internal node's minimax value from its children.
  - Max chooses a child with the highest minimax value; Min chooses one with the lowest.

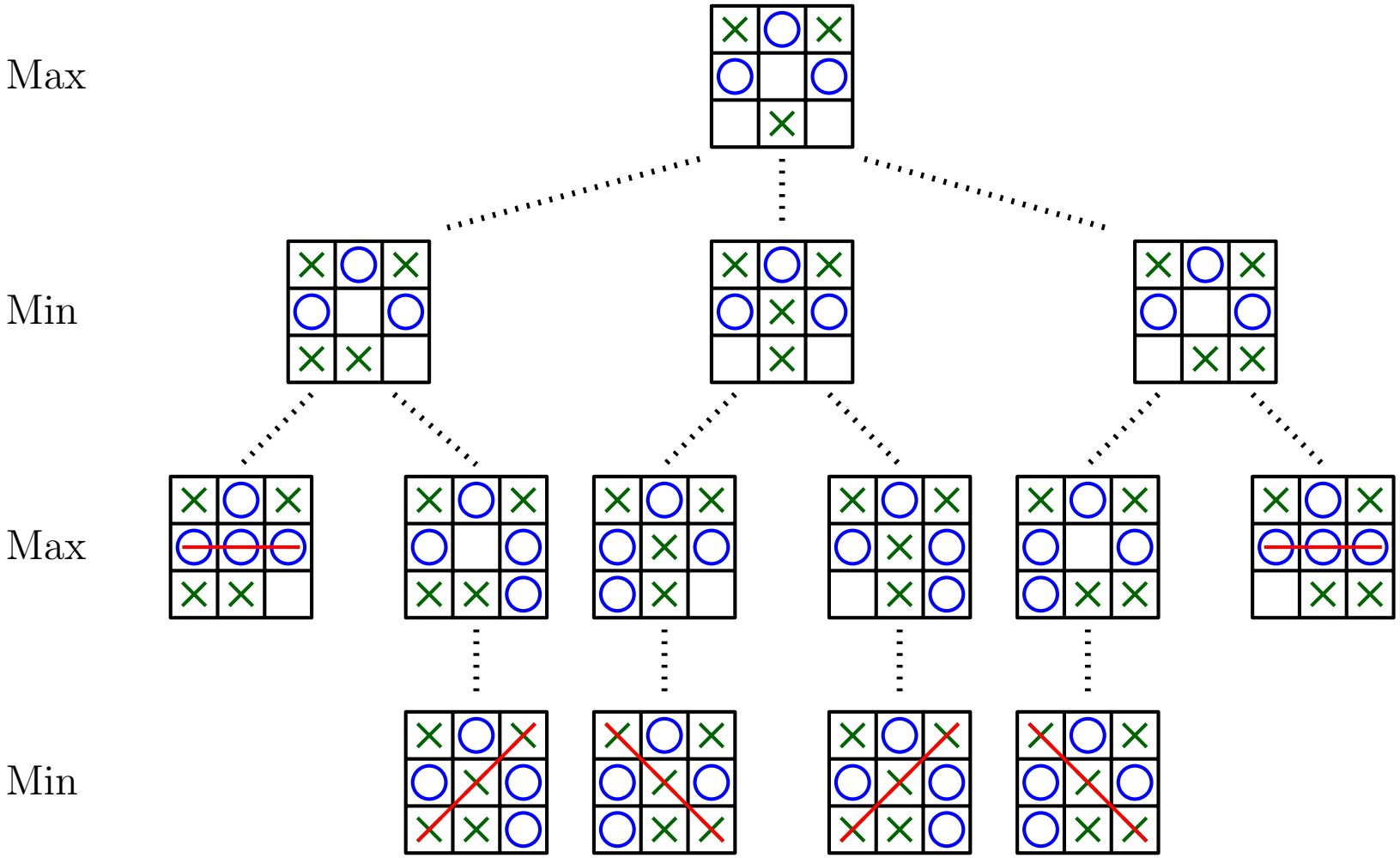
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Minimax Search: Tic-Tac-Toe Example



Example: A partial game tree starting with Max's turn. The tree alternates between Max placing × and Min placing ○, with the current player shown on the left.



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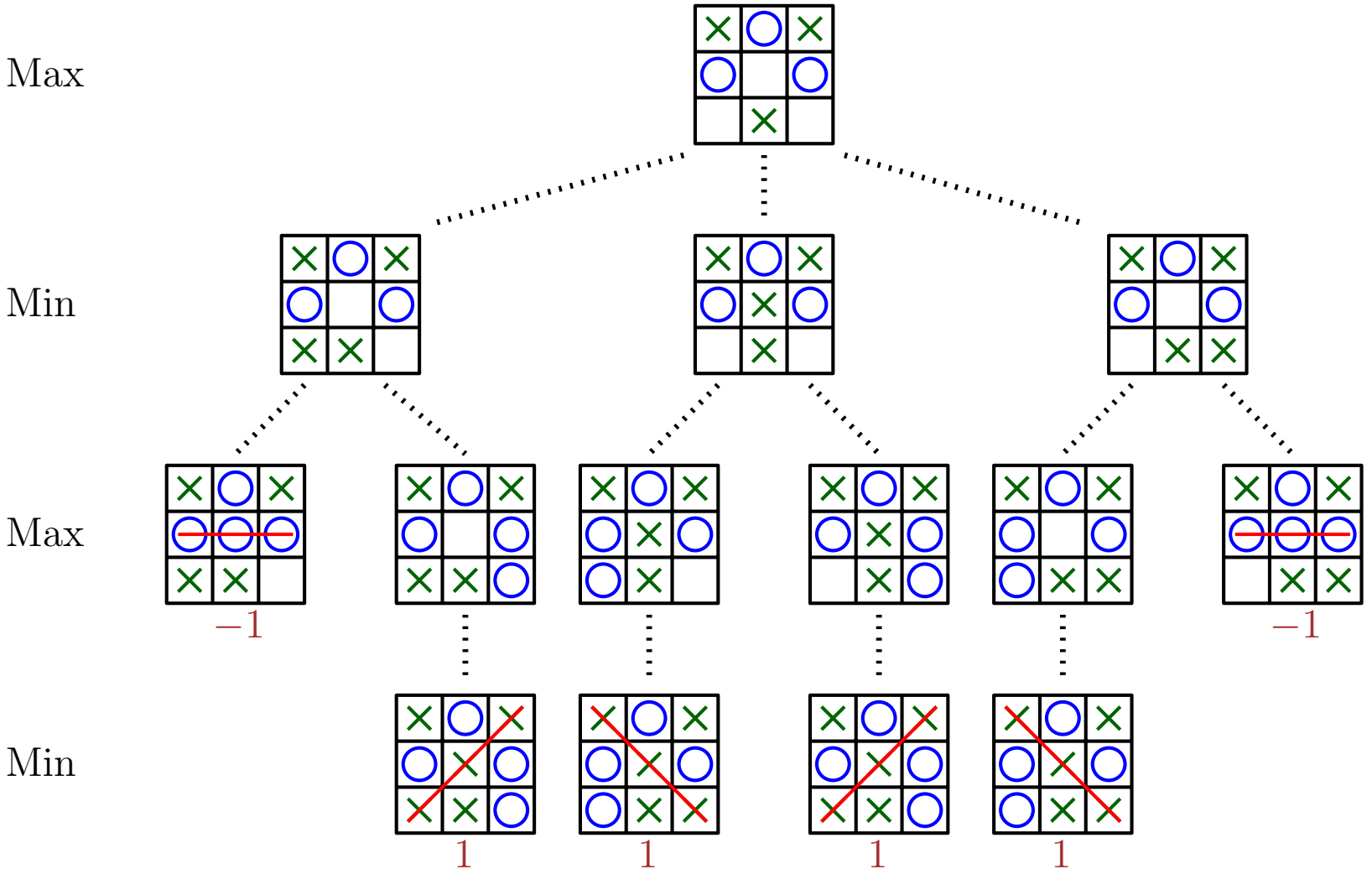
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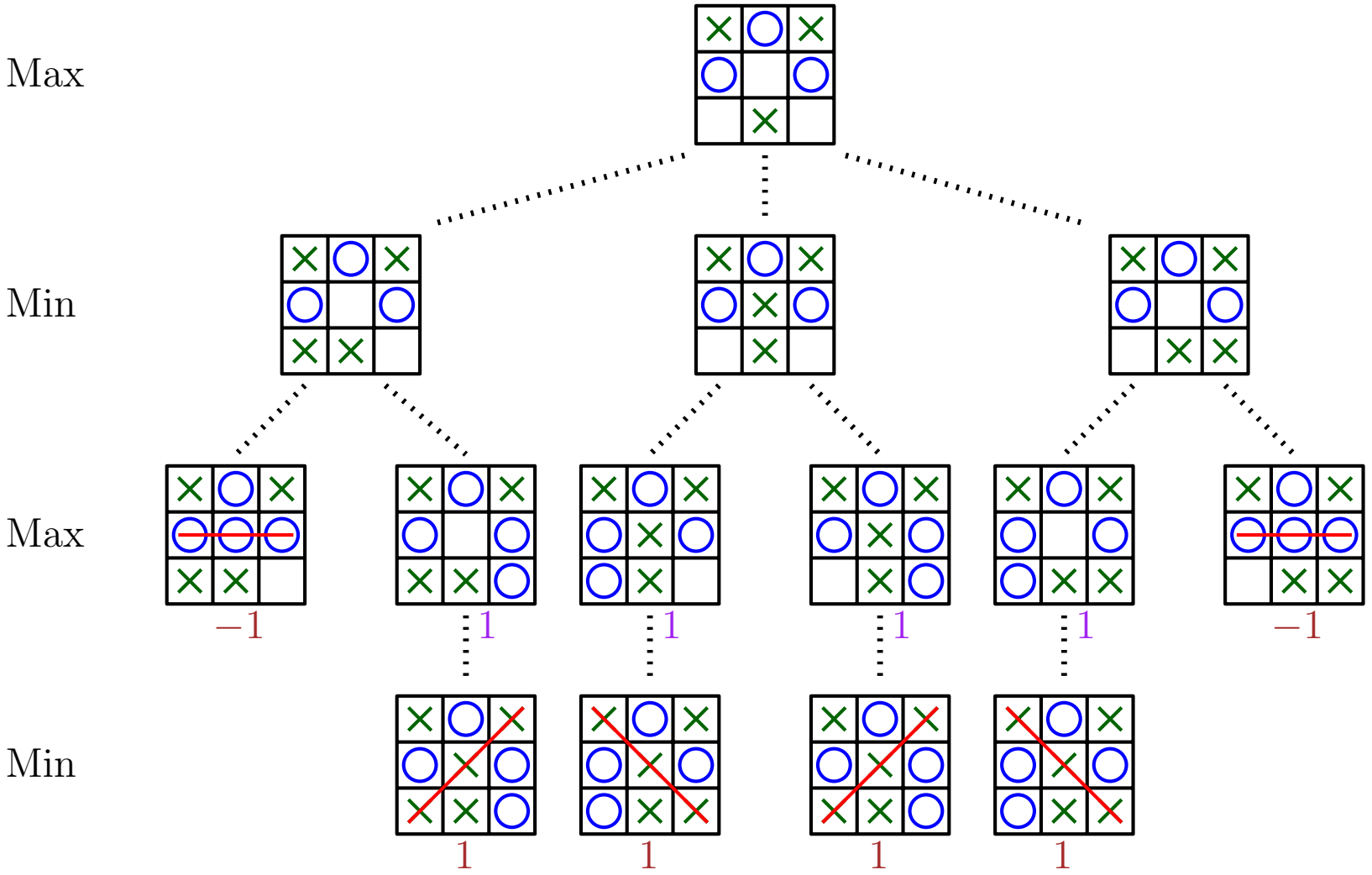


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Minimax Search: Tic-Tac-Toe Example



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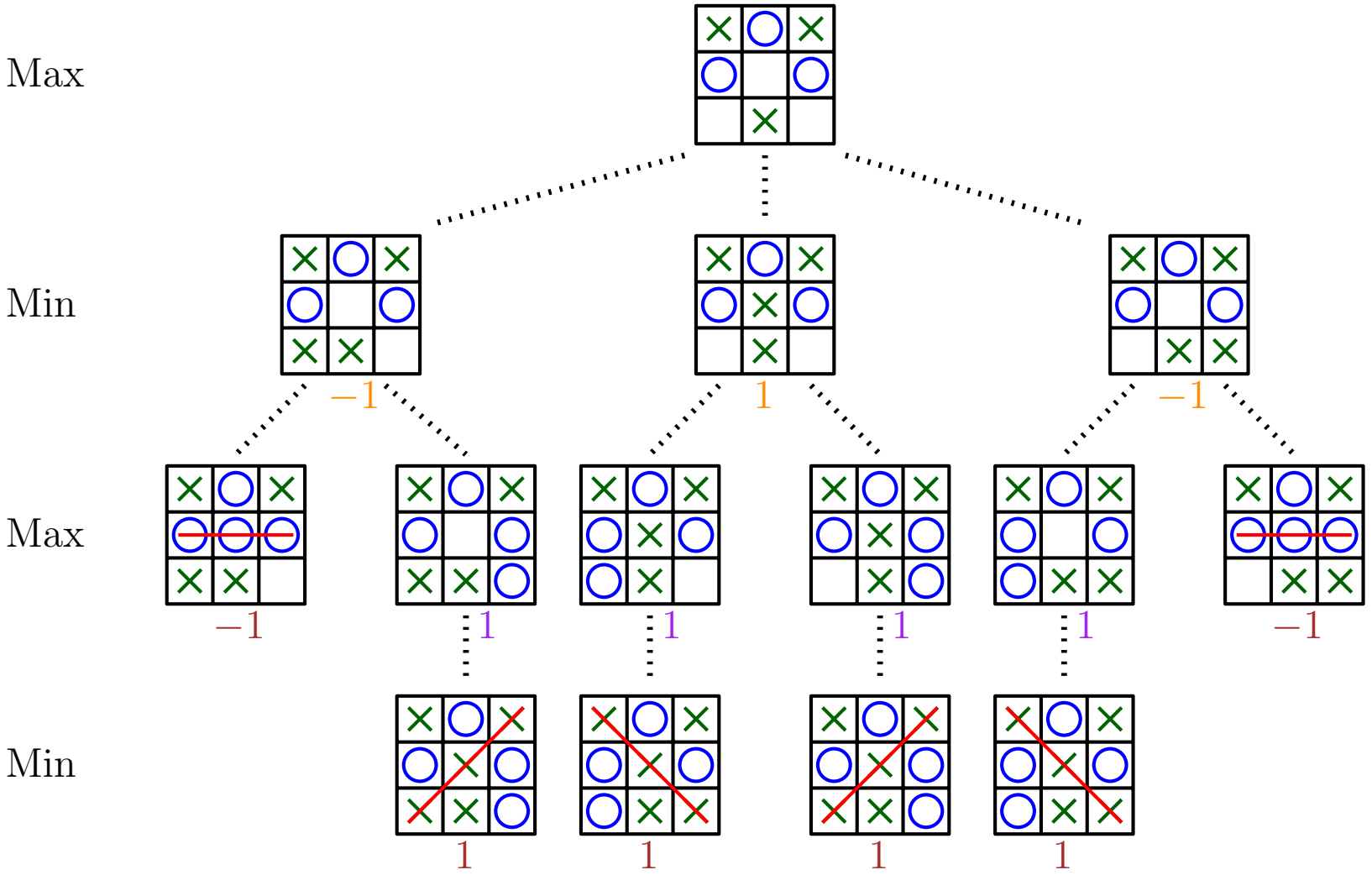
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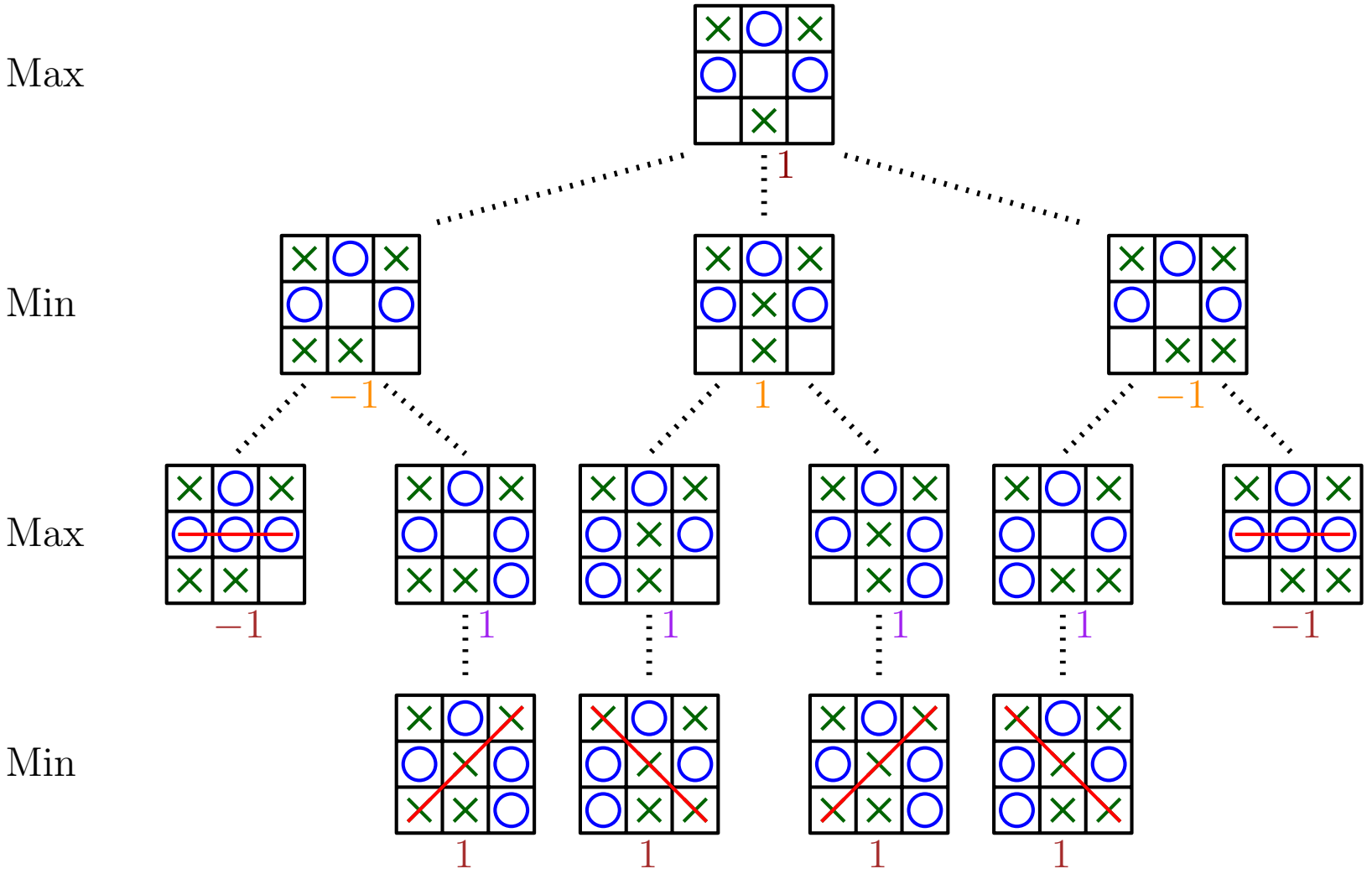


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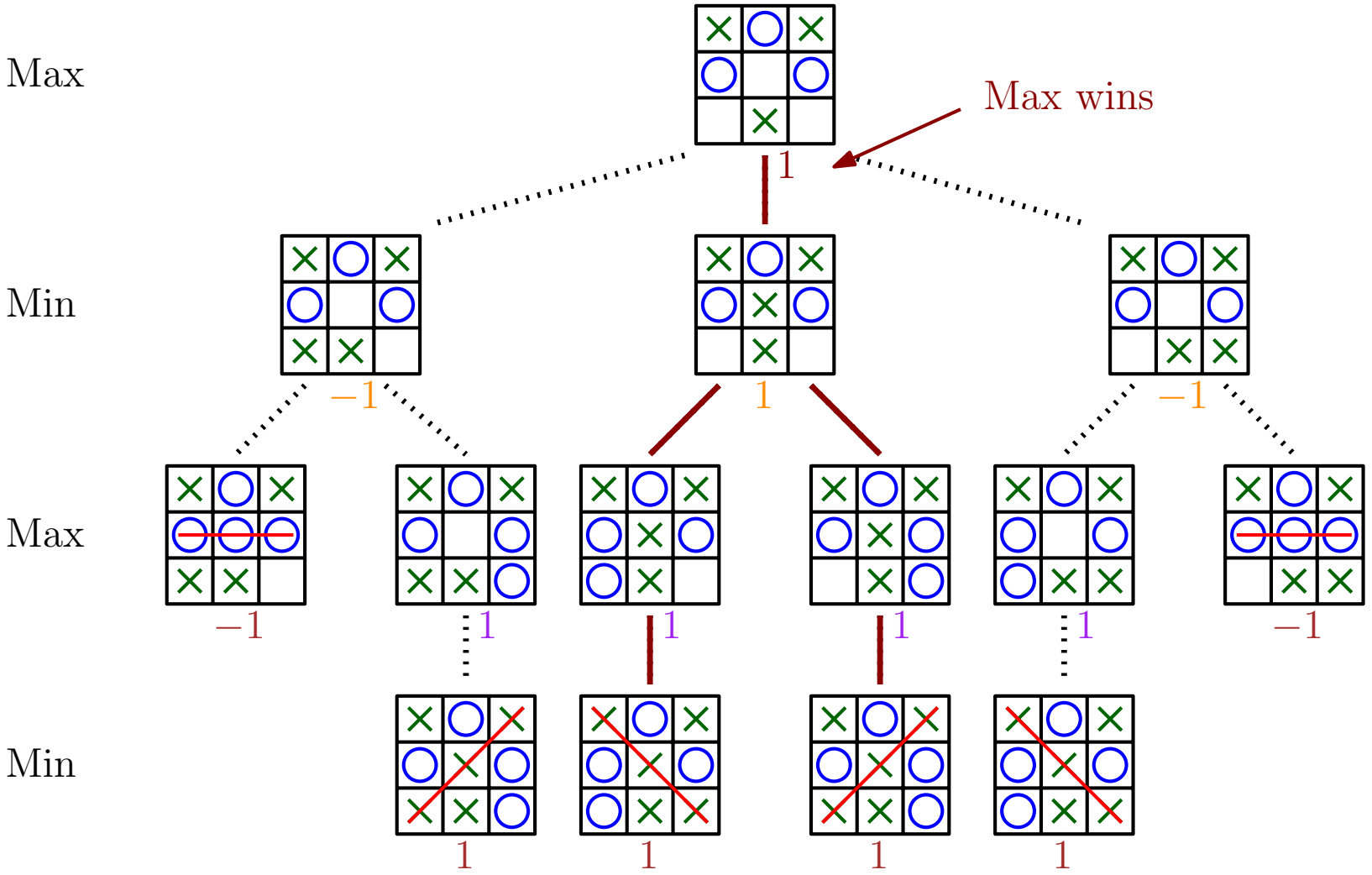
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Example: A partial game tree starting with Max's turn. The tree alternates between Max placing × and Min placing ○, with the current player shown on the left.



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Quiz

Max moves first (move A or move B); then Min replies, and the game ends. After A, Min's three replies give the scores 4, 9, and 7; after B, they give 6, 2, and 8. All scores are for Max. What is the minimax value of the root, and which move does Max choose?

(a) 9, move A

(b) 4, move A

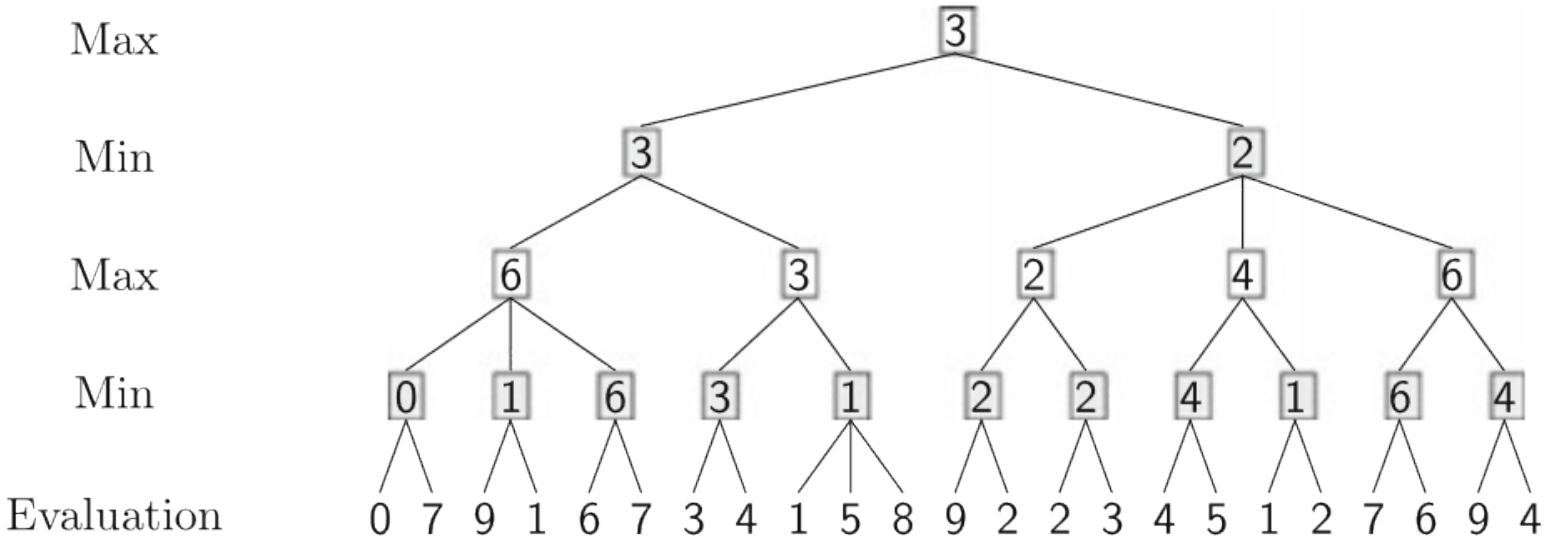
(c) 2, move B



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Exercise 1

The minimax evaluation of a game tree is shown in the figure below [Ertel 2025], Fig. 6.19, p. 117. The look-ahead is four half-moves, and the evaluations of all leaves are given. Explain how the evaluations of the inner nodes are derived.



Games with Opponents
Alpha-Beta Pruning



- **Key Idea:** Alpha-Beta Pruning avoids evaluating branches of the game tree that cannot influence the final decision based on the values already discovered during the search.
- It uses two values: *Alpha* α and *Beta* β.

Alpha and Beta Values

- **Alpha:** Represents the best (highest-value) score that the maximizing player (usually the AI) can guarantee so far. It acts as a *lower bound*. The initial value of Alpha is $-\infty$.
- **Beta:** Represents the best (lowest-value) score that the minimizing player (the opponent) can guarantee so far. It acts as an *upper bound*. The initial value of Beta is $+\infty$.

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Alpha-Beta Pruning Process



- As the tree is explored, the Alpha and Beta values are updated.
- If *Alpha* $\geq$ *Beta* at a node, its *remaining successors cannot change the root minimax value*.
- Stop examining those successors and *skip their subtrees*. The already examined path is not removed.

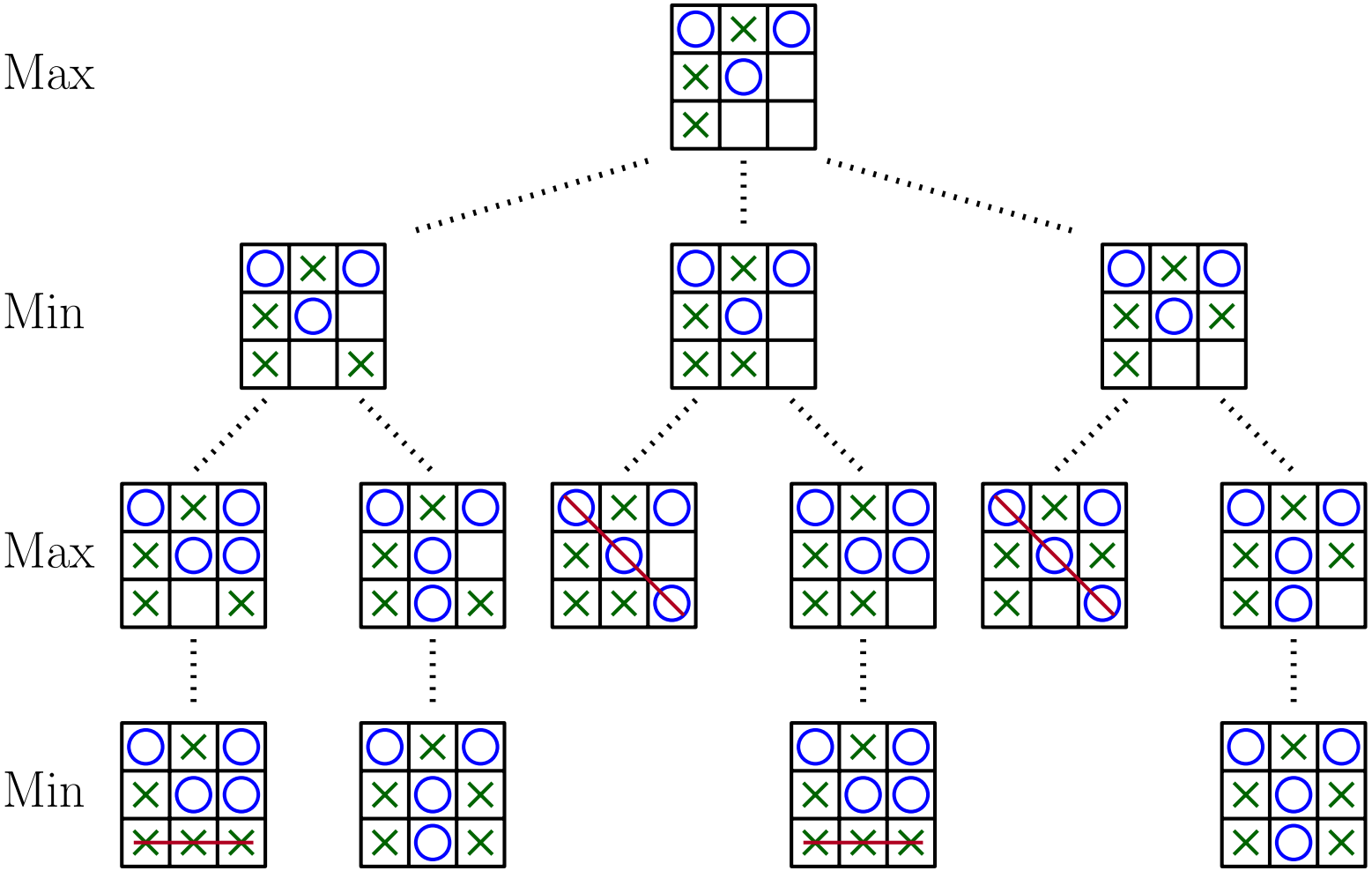
Benefits of Alpha-Beta Pruning

- Computes the same root minimax value as minimax on the same searched tree.
- Returns an optimal move while often evaluating fewer nodes.
- If several moves tie for best, the two algorithms need not select the same one.

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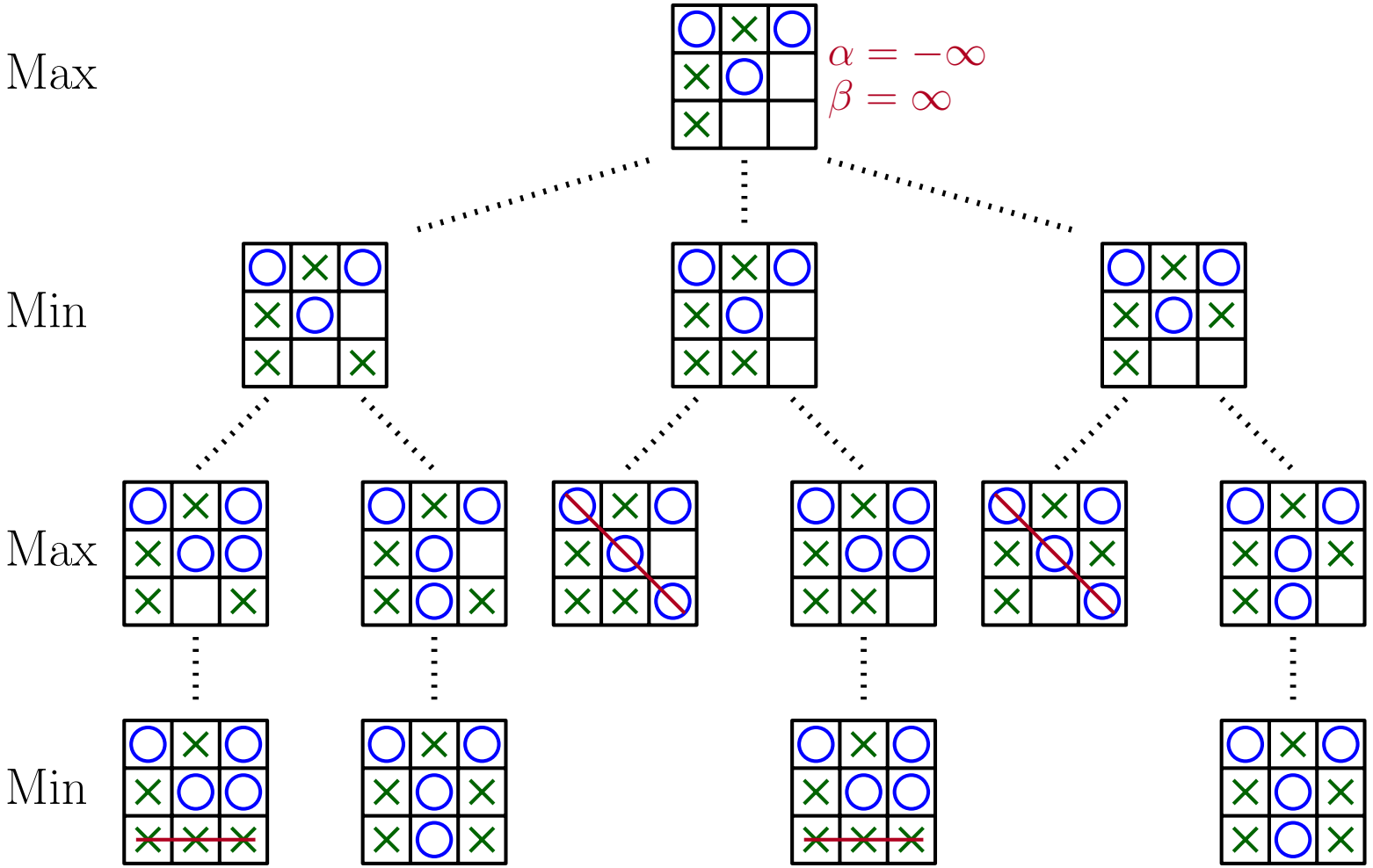


Step 1. Start at the root; search depth first from left to right.

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Step 2. Initialize the root with $\alpha = -\infty$ and $\beta = \infty$.

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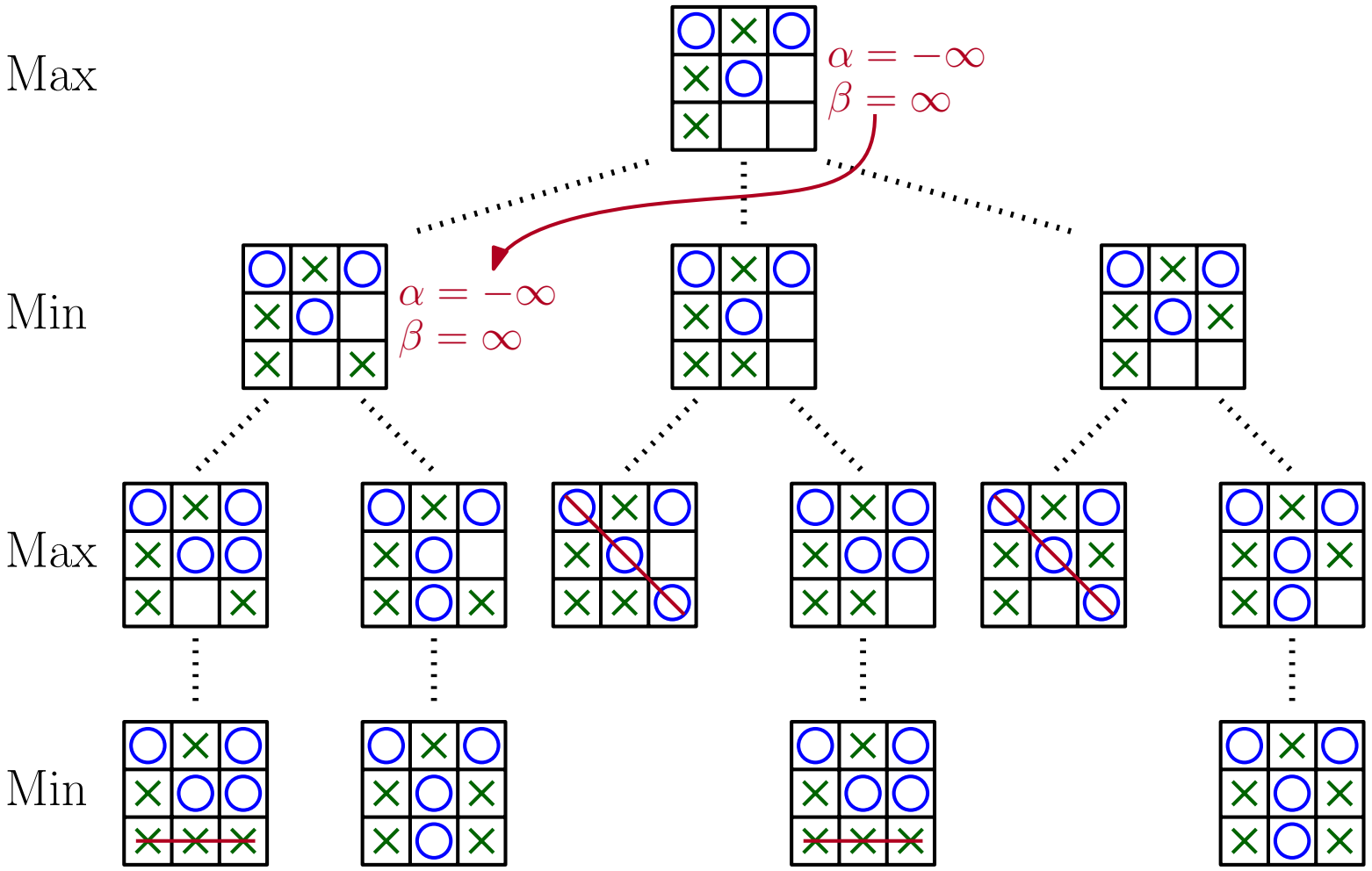
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Step 3. Visit the left Min node with $\alpha = -\infty$ and $\beta = \infty$.

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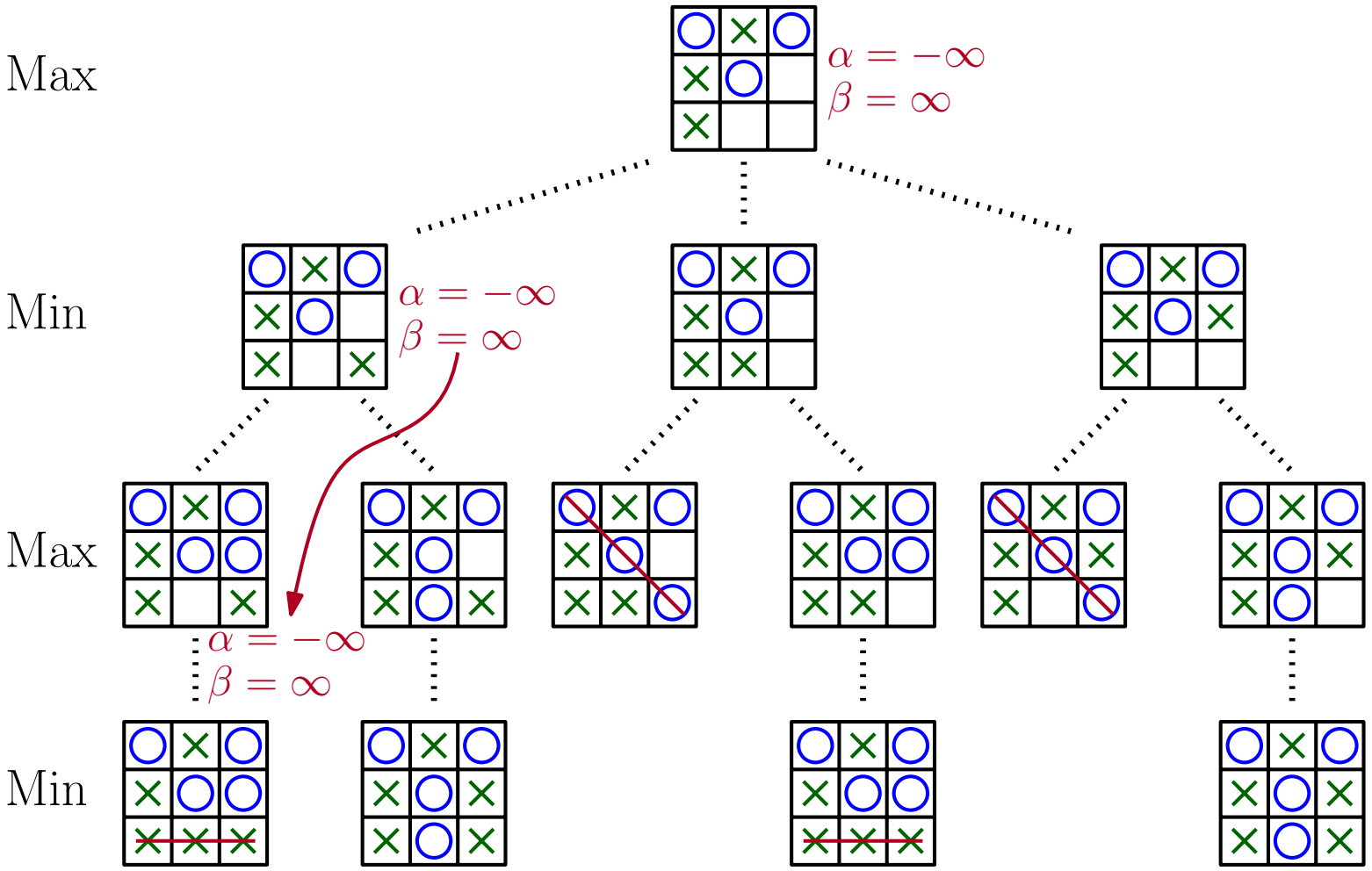
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Step 4. Visit its first Max child with the same bounds.

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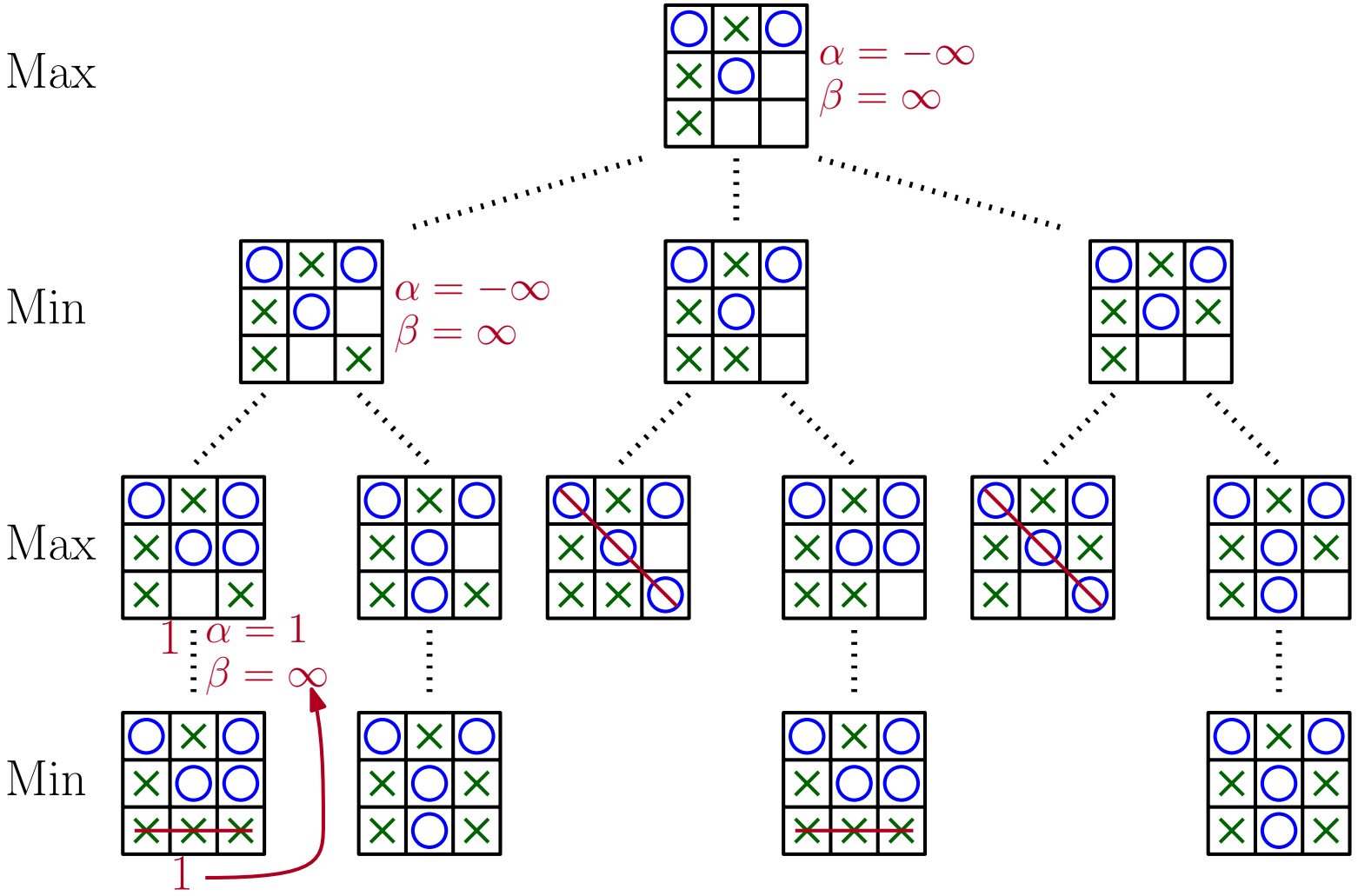
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Step 5. Terminal value 1 sets this Max node's $\alpha = 1$; return 1.

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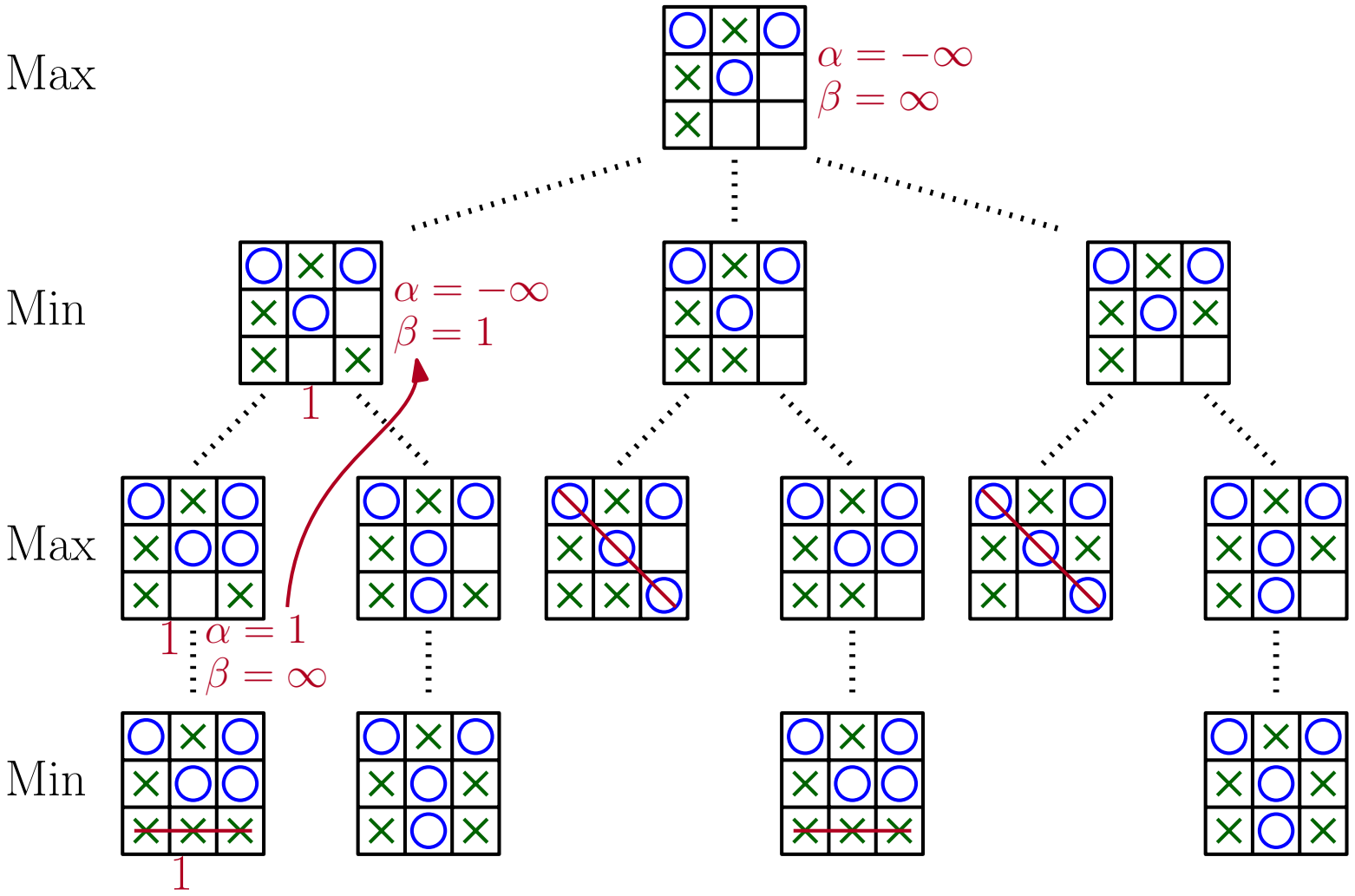
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Step 6. Left Min receives 1 and sets $\beta = \min(\infty, 1) = 1$.

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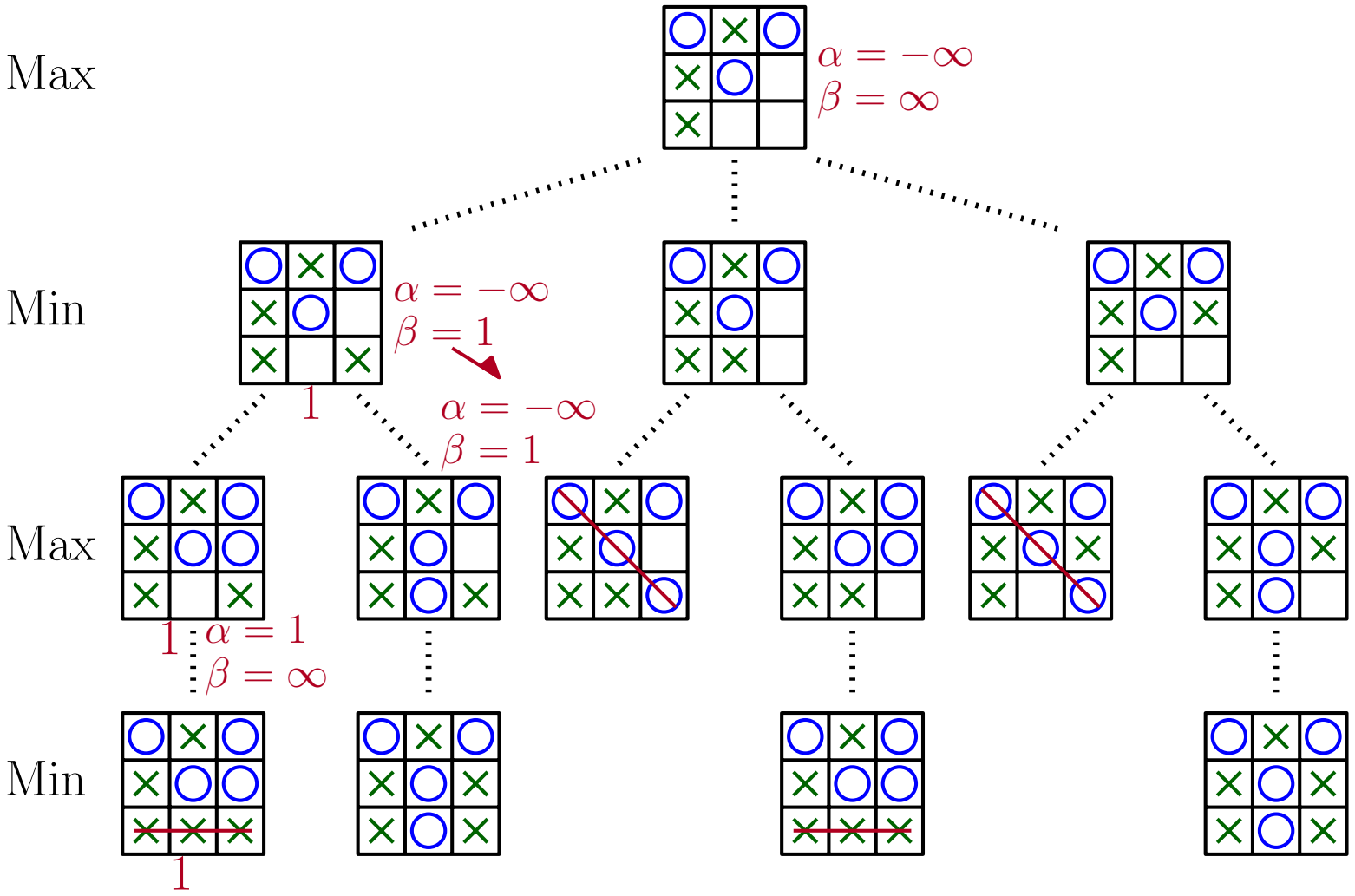
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Step 7. Visit its second Max child with $\alpha = -\infty$ and $\beta = 1$.

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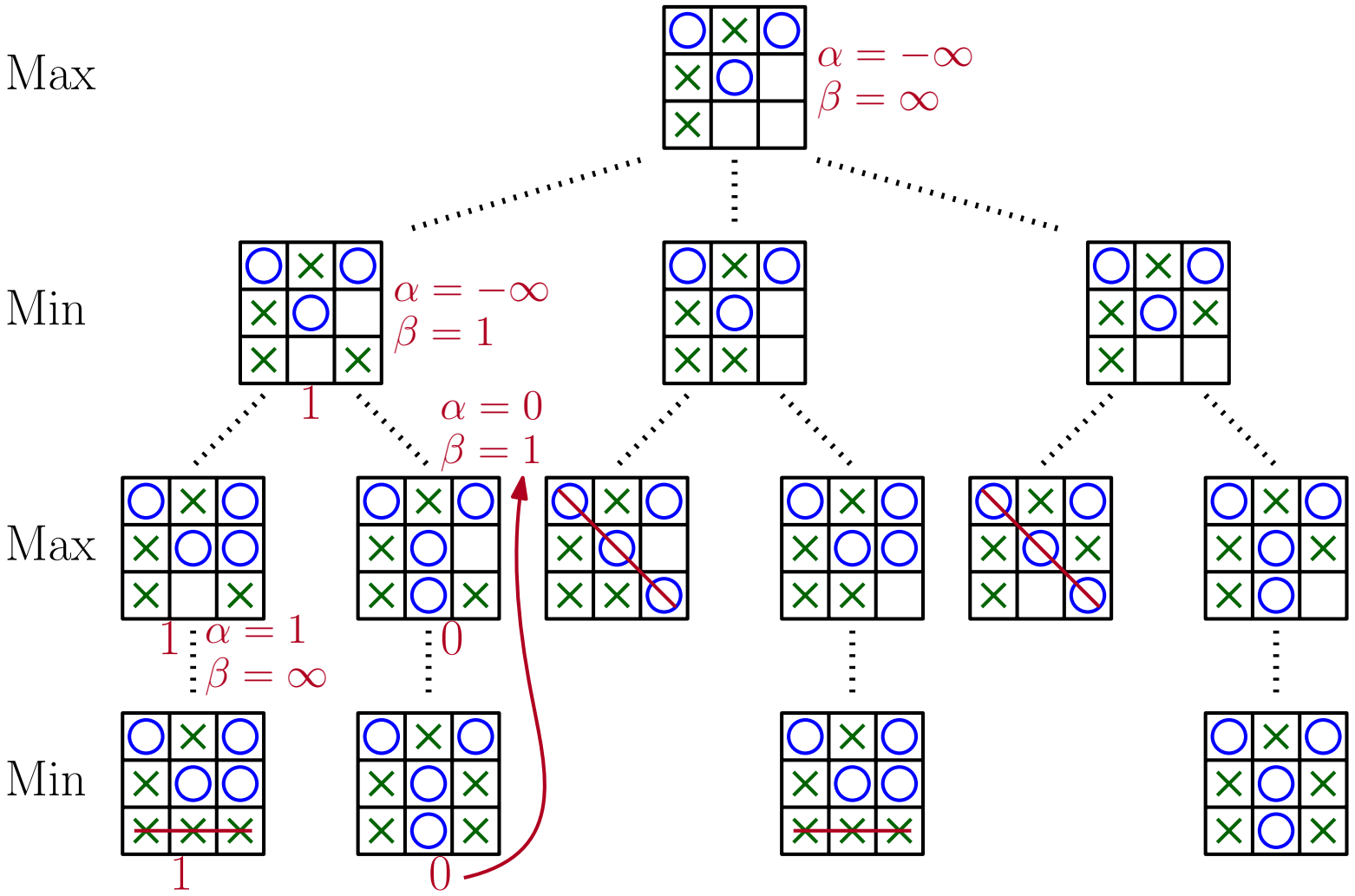
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Step 8. Terminal value 0 sets this Max node's $\alpha = 0$; return 0.

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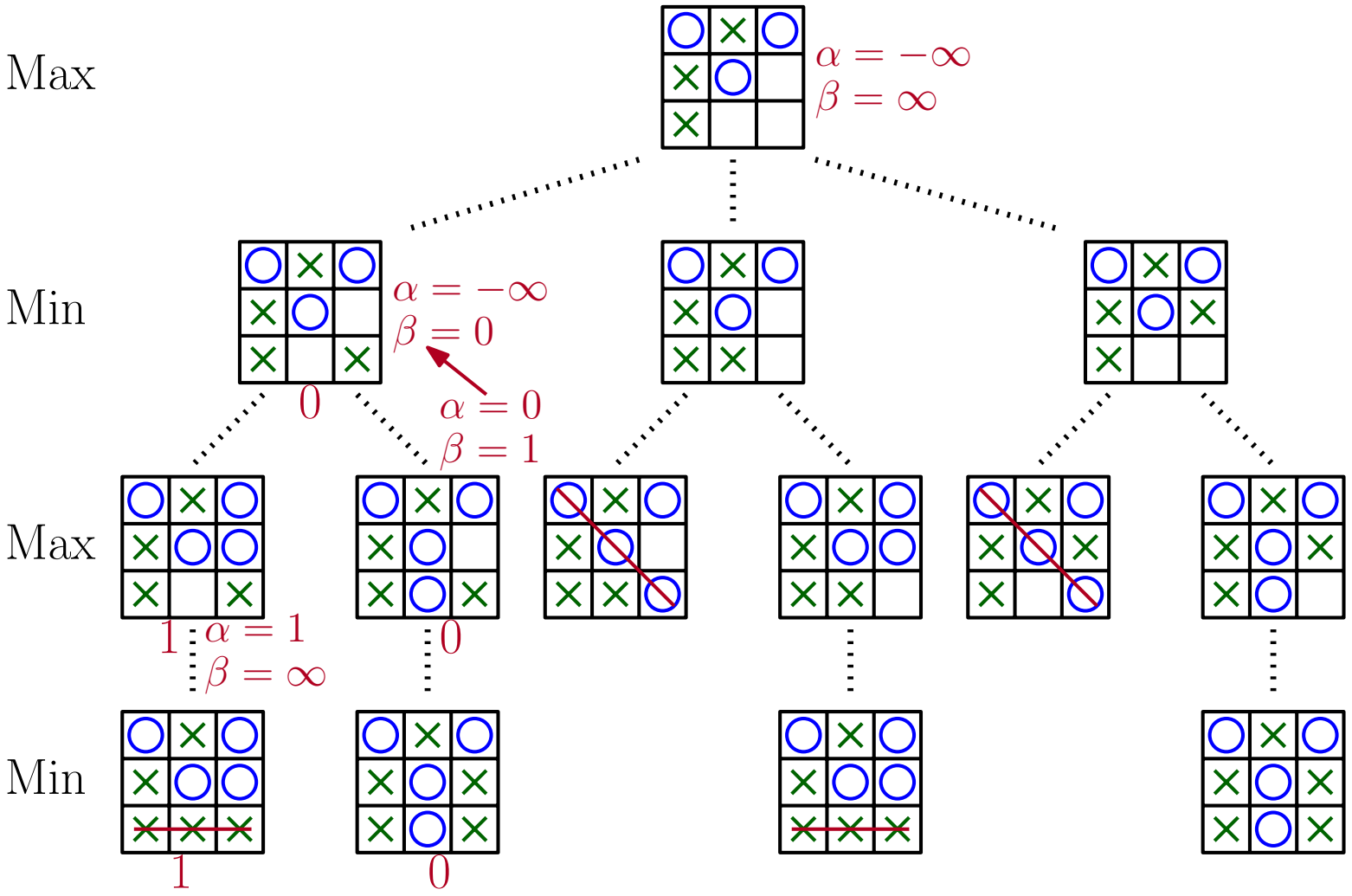
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Step 9. Left Min sets $\beta = \min(1, 0) = 0$; return 0.

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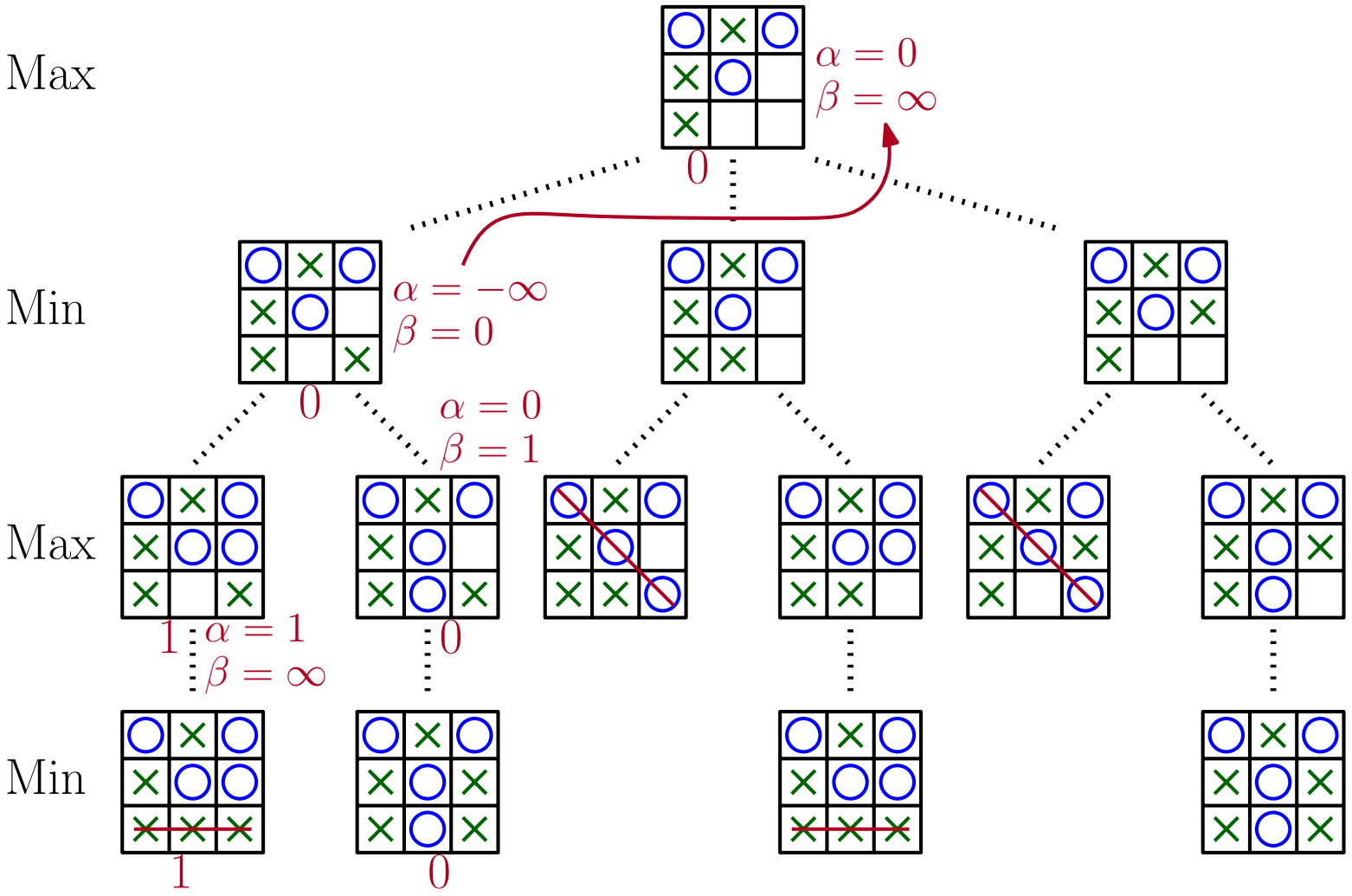
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Step 10. The root sets $\alpha = \max(-\infty, 0) = 0$.

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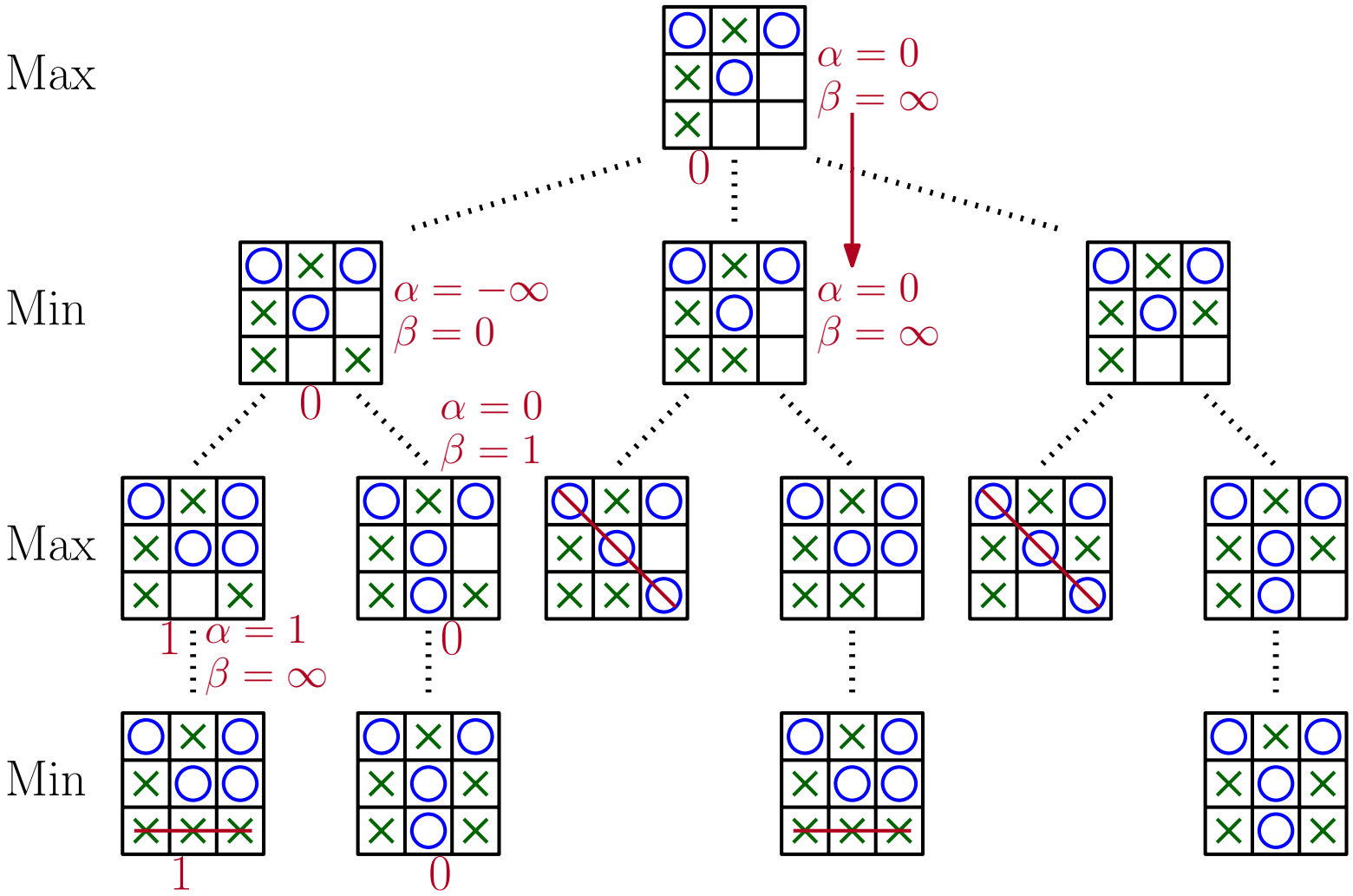
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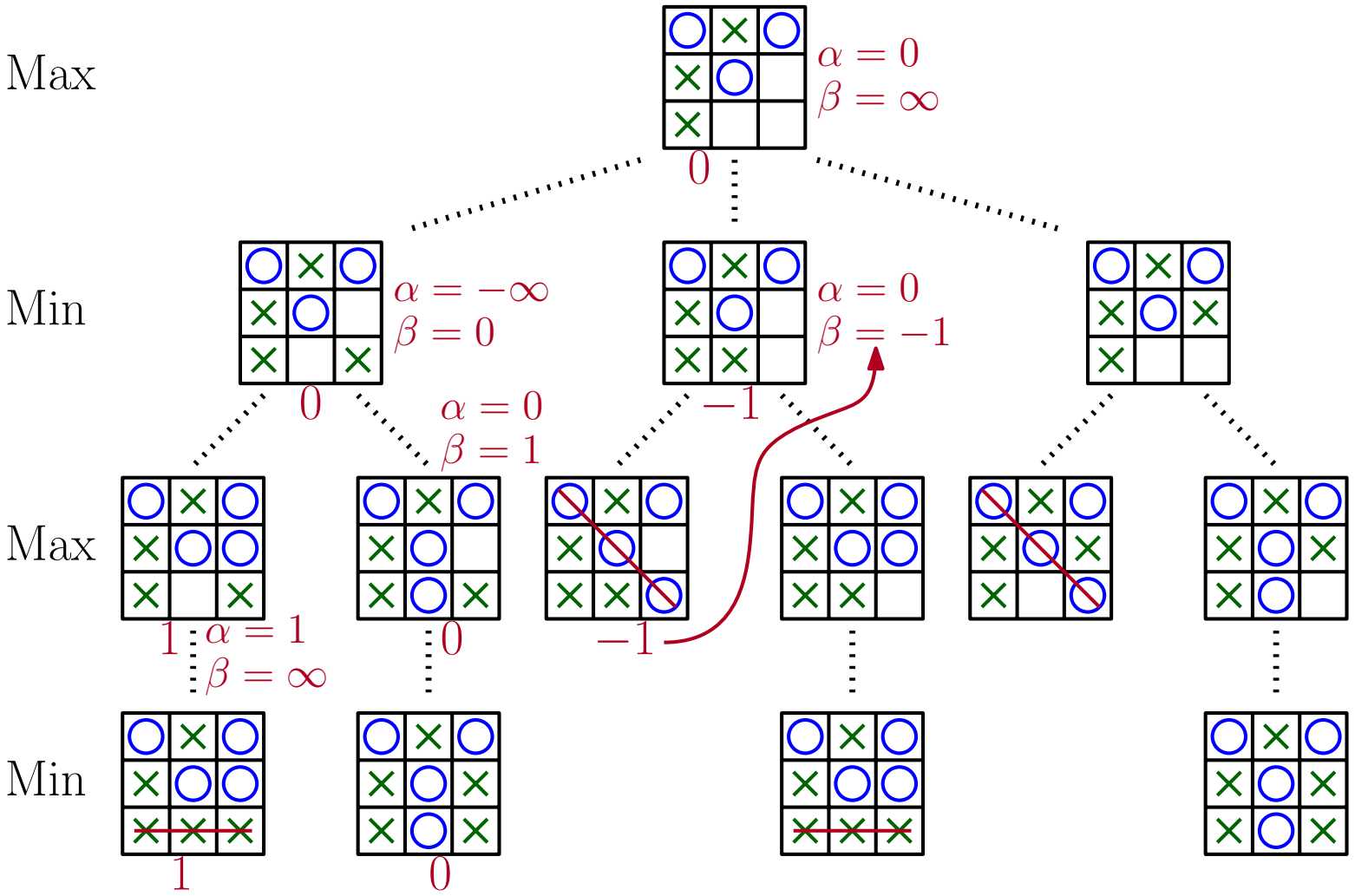
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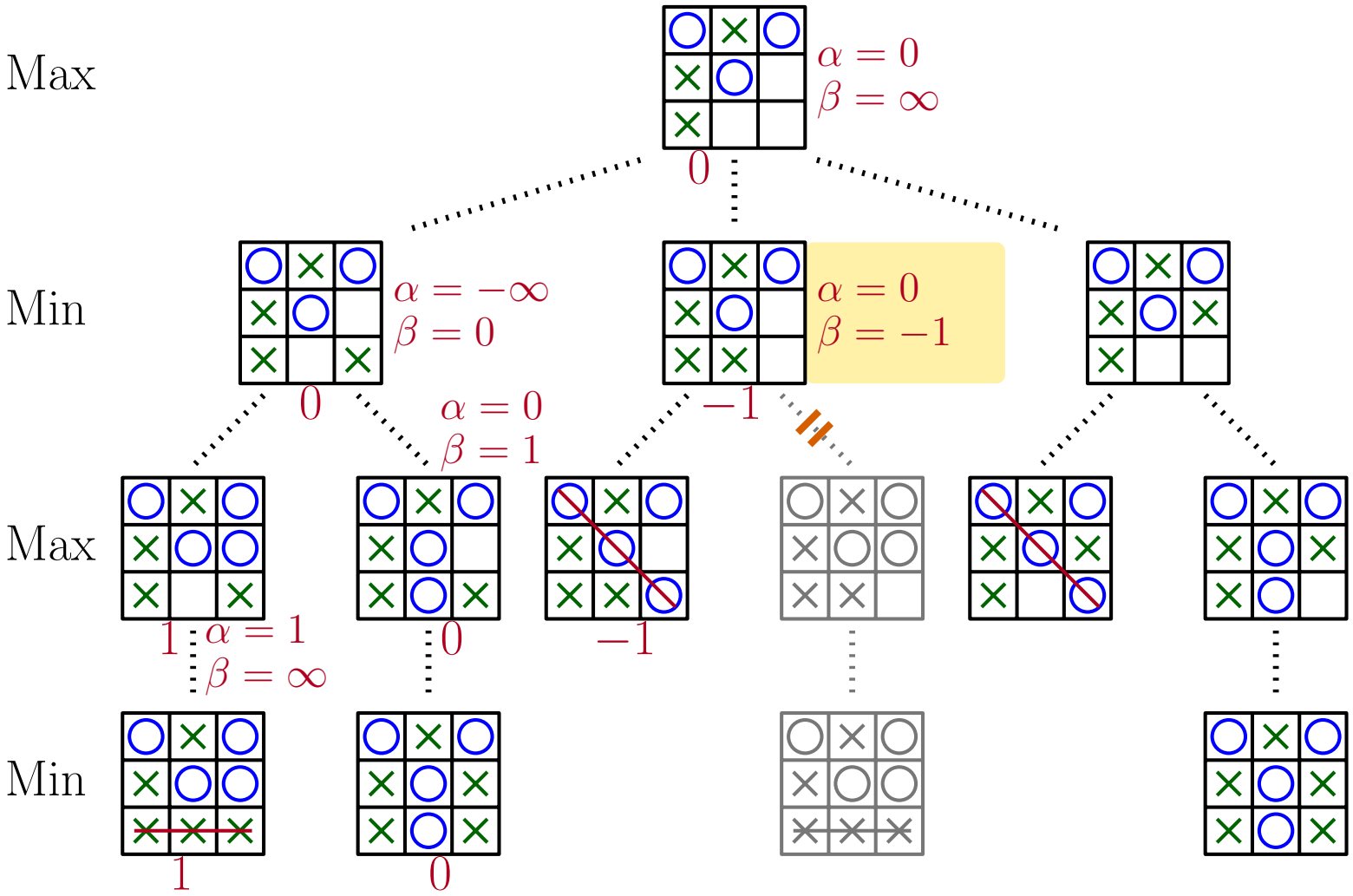
Step 11. Visit the middle Min node with $\alpha = 0$ and $\beta = \infty$.

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Alpha-Beta Pruning Process



Step 12. First value -1 sets $\beta = \min(\infty, -1) = -1$.

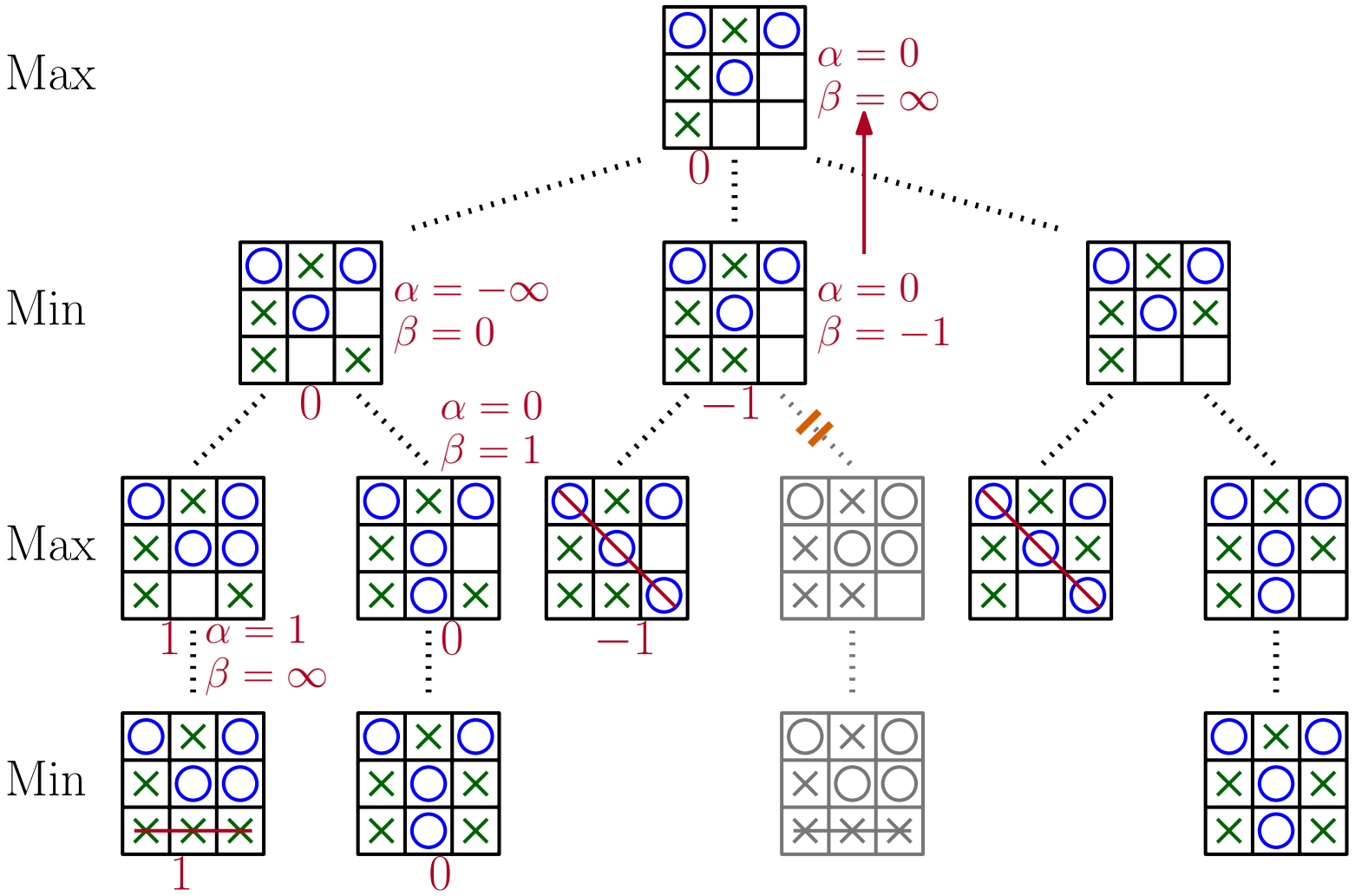
Games with Opponents
Alpha-Beta Pruning Process



Step 13. $\alpha = 0 \geq \beta = -1$, so prune the unvisited successor.

Games with Opponents

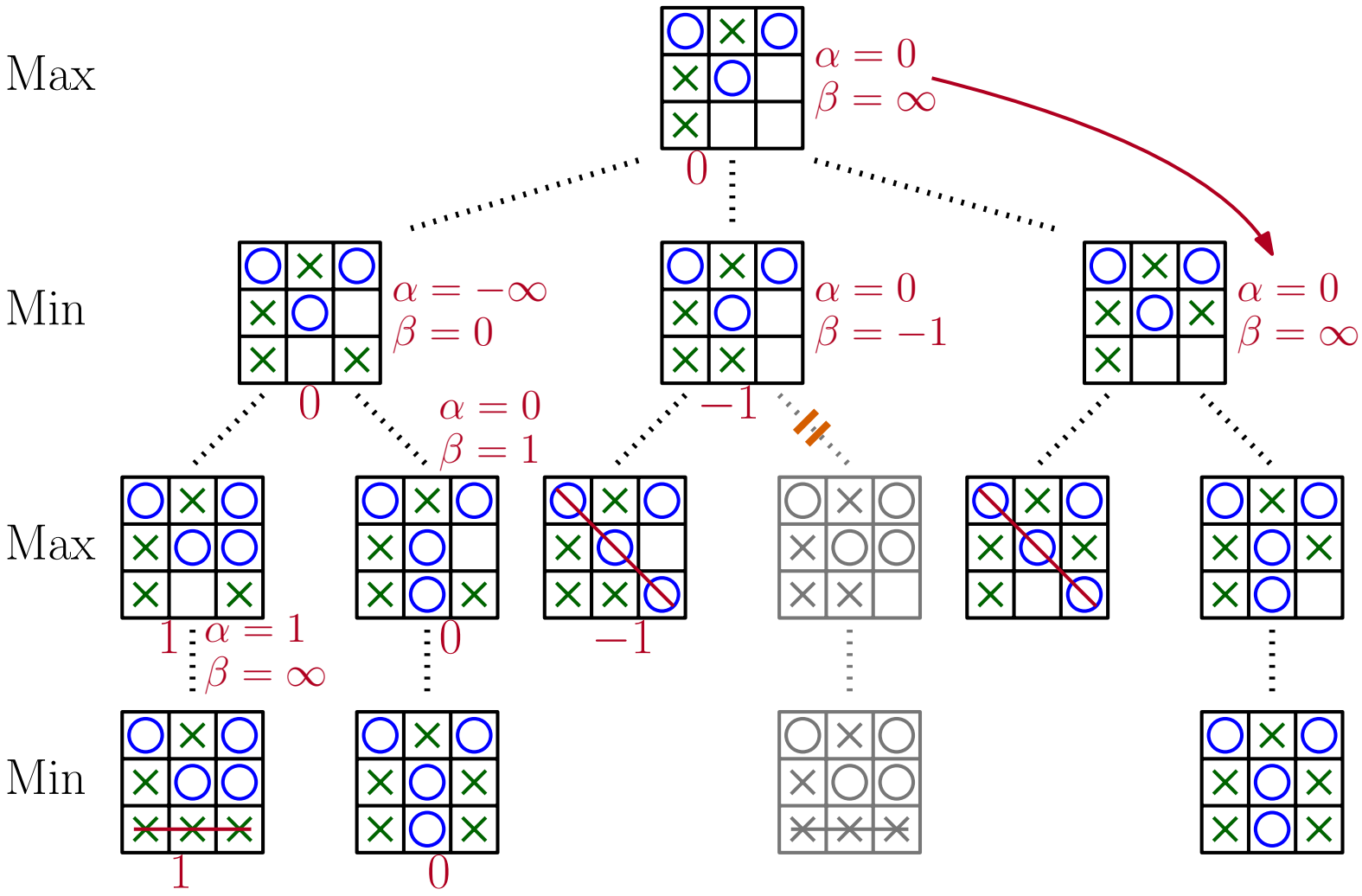
Alpha-Beta Pruning Process



Step 14. Middle Min returns -1; the root keeps 0.

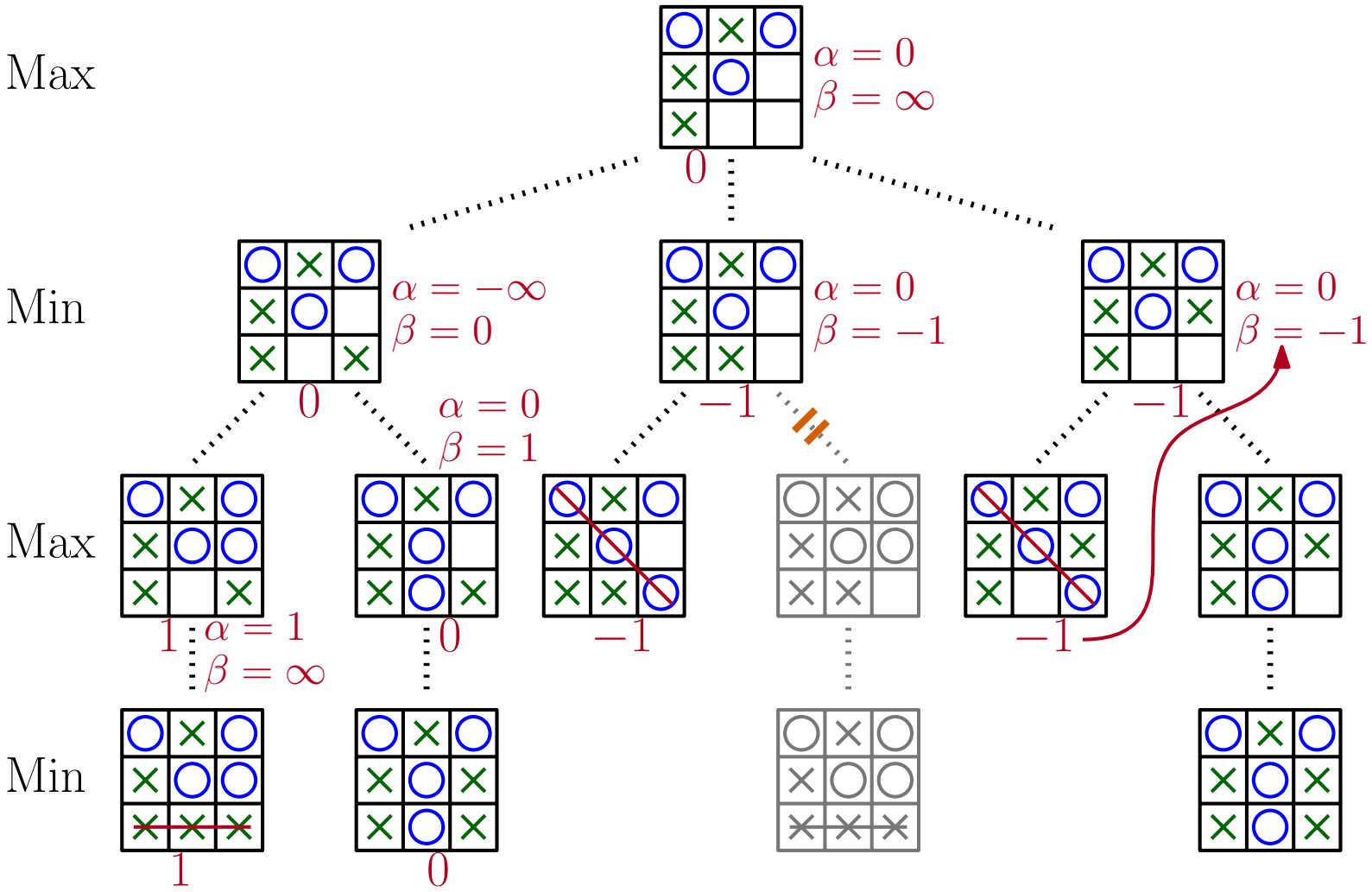
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Step 15. Visit the right Min node with $\alpha = 0$ and $\beta = \infty$.

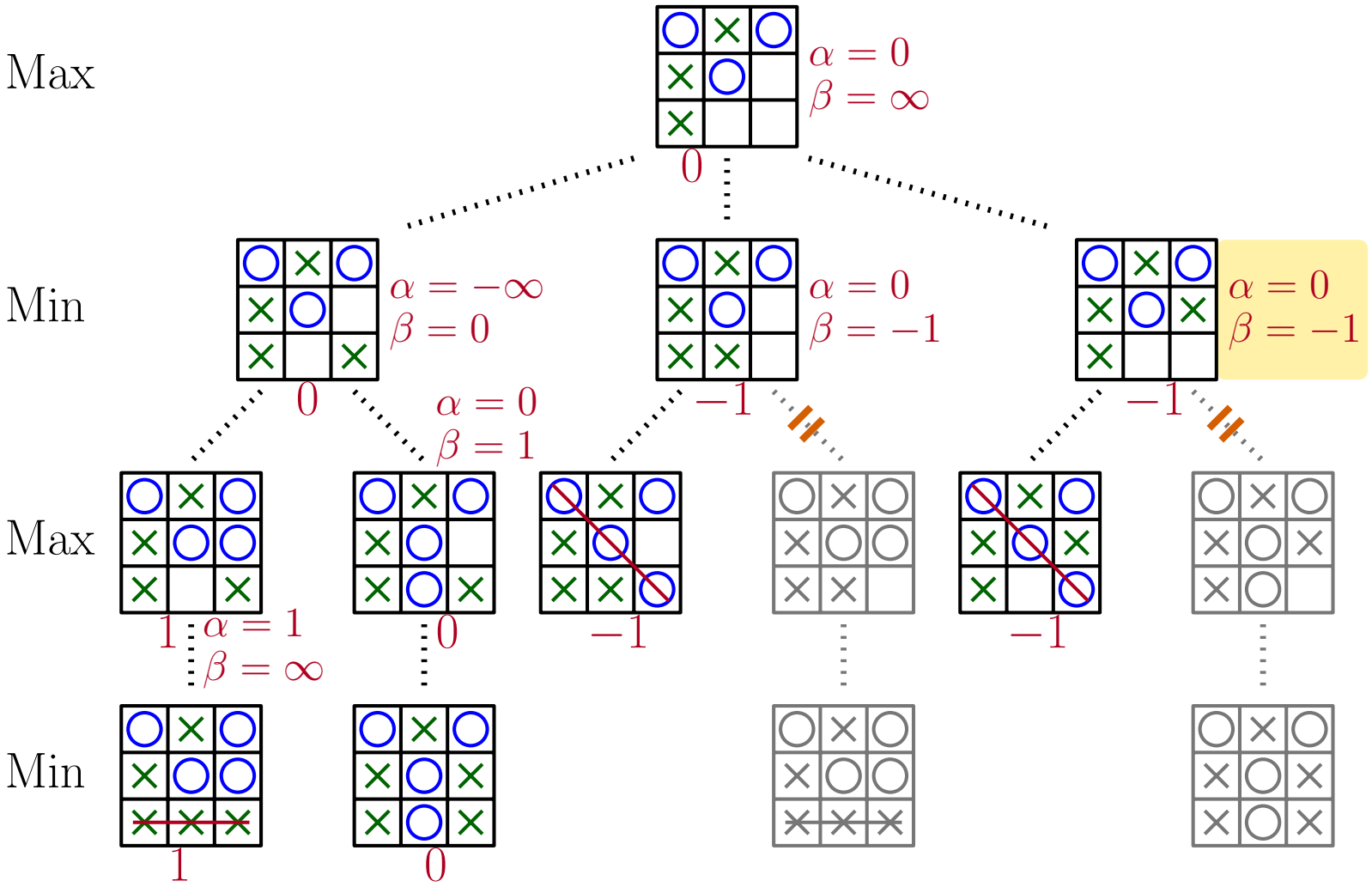
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Step 16. First value -1 sets $\beta = \min(\infty, -1) = -1$.

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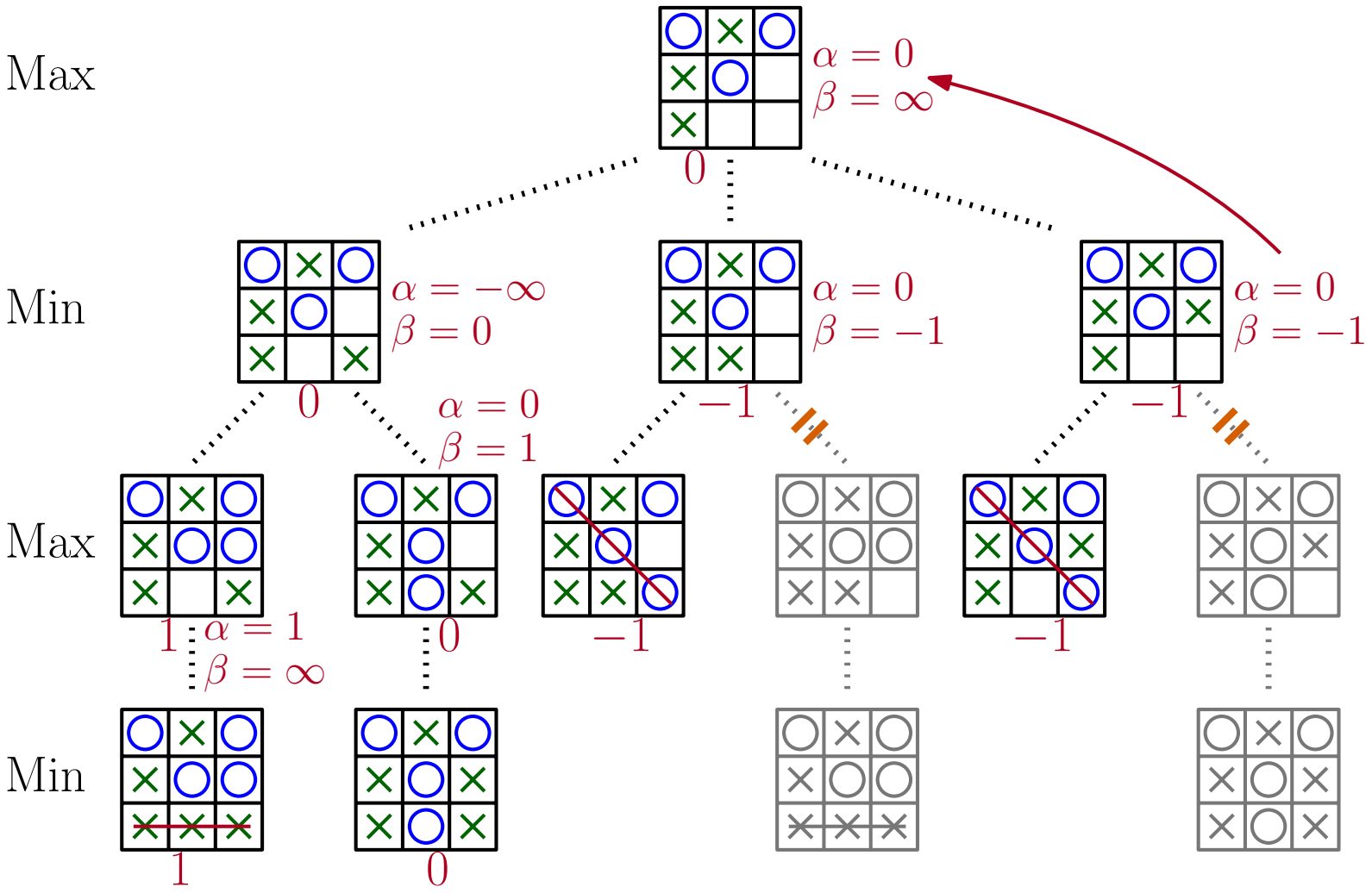
Alpha-Beta Pruning Process



Step 17. $\alpha = 0 \geq \beta = -1$, so prune the unvisited successor.

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Alpha-Beta Pruning Process



Step 18. Right Min returns -1; the root keeps its value 0.



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Quiz

Max moves first (move A or move B); then Min replies, and the game ends. After A, Min's three replies give the scores $4, 9, 7$; after B, they give $6, 2, 8$, in this order. Alpha-beta searches depth-first, from left to right. Which score is never evaluated?

(a) 9

(b) 6

(c) 8



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Quiz

At the Max root, the first Min child has been searched and has value 5. The second Min child has several children; the first one has score x. For which x must alpha-beta still look at the next child of the second Min child?

(a) $x = 2$

(b) $x = 5$

(c) $x = 7$



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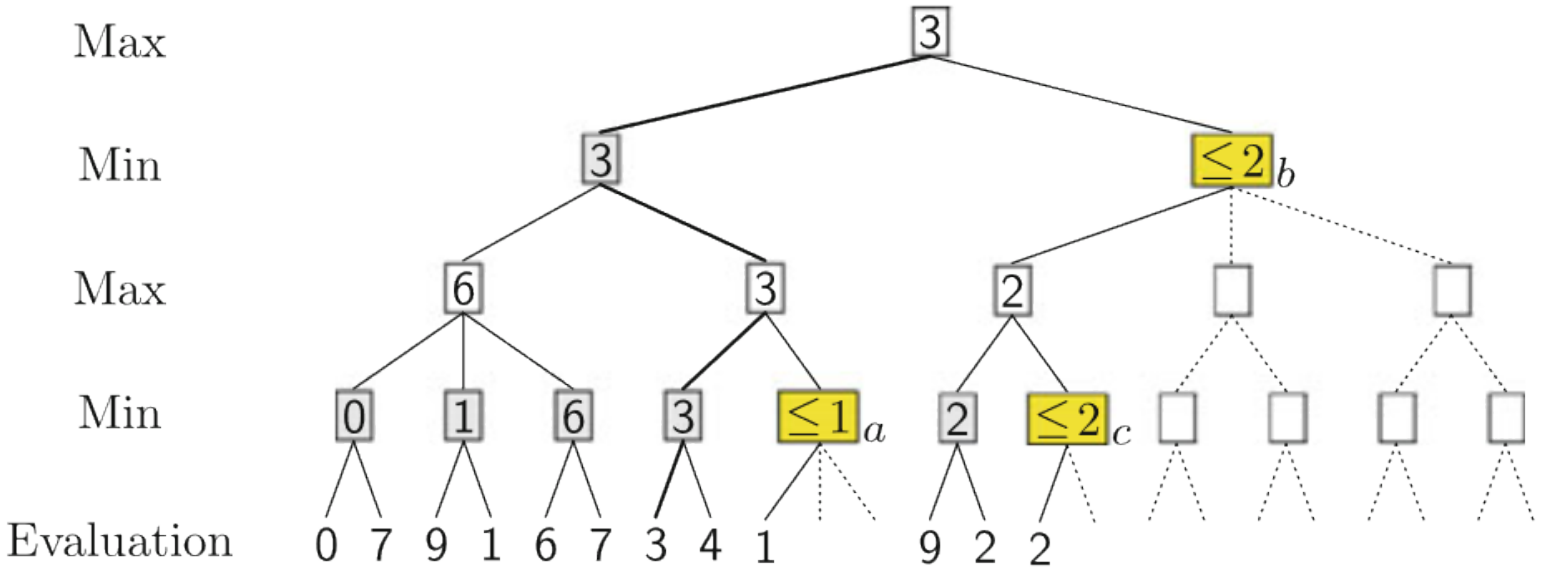
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Exercise 2

Exercise 1 gave the full game tree and all leaf evaluations from [Ertel 2025], Fig. 6.19, p. 117. The figure below shows alpha-beta pruning applied to the same tree [Ertel 2025], Fig. 6.20, p. 118. Assuming depth-first, left-to-right traversal, explain how the displayed inner-node values are obtained and why the dotted branches are not traversed.





Analysis: (see [Pearl 1984])

- *Computation time* depends heavily on *the order in which child nodes are traversed*.
- *Worst case:*
  - Visit *lower-valued successors first at Max nodes* and *higher-valued successors first at Min nodes*.
  - Alpha-beta evaluates $N_d = b^d$ leaves, the same as minimax.
- *Best case:*
  - Visit *higher-valued successors first at Max nodes* and *lower-valued successors first at Min nodes*.
  - Alpha-beta evaluates about $N_d = (\sqrt{b})^d = b^{d/2}$ leaves.
  - With the same leaf-evaluation budget, it can search *roughly twice as deep*.
  - For $b \approx 35$ in chess, the effective branching factor is about $\sqrt{35} \approx 6$.

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Quiz

A Max root has two Min children: A, with scores 4 then 5, and B, with scores 1 then 9. The scores of each child are always evaluated in this order. How many scores does alpha-beta evaluate if it searches A first, and if it searches B first?

(a) A first: 3; B first: 4

(b) A first: 4; B first: 3

(c) 3 in both cases



Analysis (cont.):

- *Average-Case:*
  - *The child nodes are randomly sorted*.
  - Effective branching factor $\approx b^{3/4} \Rightarrow N_d \approx b^{3d/4}$ leaf nodes [Pearl 1984].
  - In chess, this means effective branching factor reduces from 35 to about 14.
- *Heuristic node order:*
  - Connect alpha-beta pruning with iterative deepening over the depth limit $\Rightarrow$ *At every new depth limit we can access the ratings of all nodes of previous levels and order the successors at every branch*.
  - Effective branching factor of roughly 7 to 8, which is close to the optimal value $\sqrt{35} \approx 6$ [Ertel 2025], p. 119.

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Quiz

Let b be the branching factor. In a fixed time, minimax can evaluate about b^6 leaves, so it searches 6 half-moves deep. With children in random order, alpha-beta evaluates about $b^{3d/4}$ leaves at depth d. About how deep can alpha-beta search in the same time?

(a) 6 half-moves

(b) 8 half-moves

(c) 12 half-moves

Games with Opponents
Non-deterministic Games



- Consider a game in which *one fair six-sided die* is rolled before each move.
- The game tree contains *Max*, *Min*, and *chance nodes*, in the repeating order Max, chance, Min, chance, . . .
- A chance node has *one successor for each possible die outcome*.
- Its value is the *probability-weighted average* of the resulting position values. Because all six outcomes have probability $1/6$ here, this is the ordinary average of the six successor values [Russell and Norvig 2021], Sec. 5.5, pp. 165–166.

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Quiz

Max chooses move A or move B; then a fair die is rolled. After A, the six outcomes are worth $3, 3, 3, 3, 3, 3$; after B, they are worth $0, 0, 0, 6, 6, 12$. Using the chance-node rule (probability-weighted average), what are the values after A and after B, and which move does Max choose?

(a) A: 3, B: 0; Max chooses A

(b) A: 3, B: 12; Max chooses B

(c) A: 3, B: 4; Max chooses B

Heuristic Evaluation Functions

From Expert Knowledge to Features



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The following example illustrates how to construct a heuristic evaluation function using human expertise [Ertel 2025], Sec. 6.5, p. 120.

Example 8

- Experts identify the *most important factors for selecting a move*. These factors are quantified as *features or attributes*.
- In the simplest case, the features are combined into a *linear evaluation function* $B(s)$ for a position s:

$$\begin{aligned} B(s) = a_1 \cdot \text{material} + a_2 \cdot \text{pawn_structure} \\ + a_3 \cdot \text{king_safety} + a_4 \cdot \text{knight_in_center} \\ + a_5 \cdot \text{bishop_diagonal_coverage} + \cdots \end{aligned}$$



Example 8 (cont.)

- One important feature is the *material advantage*, measured from *Max's perspective*:

$$\begin{aligned}\text{material} &= \text{material}(\text{Max}) - \text{material}(\text{Min}),\\ \text{material}(P) &= 100 \cdot \text{num_pawns}(P) + 300 \cdot \text{num_knights}(P)\\ &\quad + 300 \cdot \text{num_bishops}(P) + 500 \cdot \text{num_rooks}(P)\\ &\quad + 900 \cdot \text{num_queens}(P).\end{aligned}$$

- Initially, the weights a_i can be set using *expert judgment* and adjusted after games. A *machine-learning method* can optimize them more systematically.



Example 8 (cont.)

- Human experts may identify relevant features $f_1(s), f_2(s), \ldots$, while a *learning algorithm adjusts how the features are combined*.
- A simple learning loop is:
  - play games using the current evaluation function;
  - observe the delayed result: victory, defeat, or draw;
  - adjust the evaluation function to improve future decisions.
- However, the final result does not reveal *which earlier decisions contributed to it* or how the evaluation function should change. This is the *credit-assignment problem*.
- *Reinforcement learning* addresses such delayed feedback.
- *Stockfish combines learning and search*: a *trained neural network evaluates chess positions*; during a game, alpha-beta search uses these scores to choose a move [Stockfish developers 2020].

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Take-Home Messages



- Early game-playing programs combined *limited search* with *human-designed evaluation*.
- Strong classical chess systems used *larger alpha-beta searches* and increasingly refined evaluation functions.
- *Monte Carlo tree search* introduced a different approach: spend more computation on promising continuations.
- Modern systems learn *evaluation or search guidance* from games or self-play and combine it with search; *search remains central*.

Search, Evaluation, and Learning in Games



Exercise 3

Choose one chess engine or research system released or substantially updated within the last five years. In 300–500 words:

- identify the system, version, and release or publication date;
- explain how it uses search, position evaluation, and learning;
- report one performance result and name the metric or benchmark;
- compare one design choice with an earlier system discussed in this section.

Use at least two reliable sources, including one primary source such as a research paper, official technical document, or official source repository. Give complete citations and state when the information was last checked.

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References

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